



SHZV Type Vacuum On-load Tap-Changer

Technical Data

HM 0.154.3901-09.25/2024

*Caution:
For safe and proper use,
follow these instructions.
Retain for future reference.*



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1. General

Type SHZV vacuum on-load tap-changer (herein referred to as tap-changer) is of combined structure, applicable to oil-immersed voltage regulating transformers. The tap-changer is composed of a diverter switch and a tap selector, immersed in transformer oil. Diverter switch is in a separate oil compartment. Tap selector is in the same transformer oil tank as the windings. Tap-changer is to be mounted on transformer tank top by means of bell type mounting through a top flange. Tap-changer is operated by a motor drive unit. Tap-changer and motor drive unit are connected by an upper gearbox, driving shaft and a bevel gearbox. Tap-changer provides both local and remote operation modes.

Tap-changer connection includes neutral point of star connection for three-phase and any connection for single phase, three units of single phase tap-changers can be used for any selectable winding connections for a three phase transformer.

Its basic connection diagram is shown in Fig. 1-1 below:

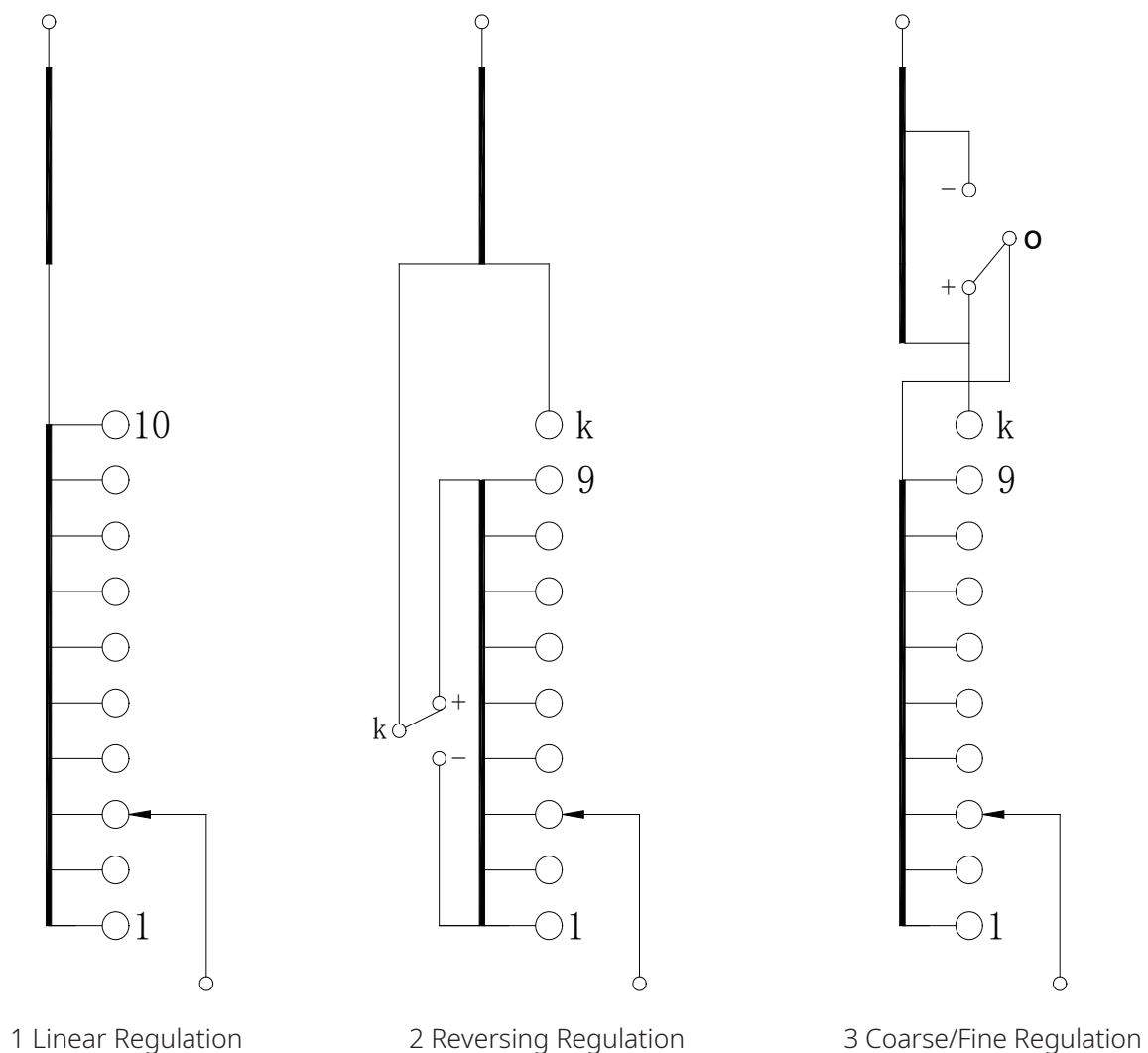


Fig.1-1 Basic Connection Diagram of Tap Winding

2. Technical Specifications

Type SHZV OLTC complies with GB/T 10230.1-2019 and IEC 60214-2014 standard. Tap-changer technical data is listed in Table 2-1 below.

Table 2-1 Tap-Changer Technical Data

Description		SHZV III			SHZV I				
Max. rated through current (A)		400	600	1000	400	600	1000	1600	2400
Rated frequency (Hz)		50 or 60							
Connection		3-phase Y-connection			Single-phase any connection				
Max. rated step voltage (V)		4000							
Rated step capacity (kVA)		1500	1600	3000	1500	1600	3000	4400	5600
Short-circuit current test (kA)	Thermal (3s)	6	8	12	6	8	12	20	24
	Dynamic (peak)	15	20	30	15	20	30	50	60
Max. operating positions		1. Max. 18 without change-over selector; max. 35 with change-over selector.							
		2. Current ≤600A, selector is the same as the type CM, and max. 107 are allowed.							
Insulation to ground (kV)	The highest voltage for equipment	72.5	126	170	252	300	363		
	Rated separate source AC withstand voltage (50Hz, 1min)	140	230	325	460	480	510		
	Rated lightning impulse withstand voltage (1.2/50μs)	350	550	750	1050	1100	1175		
Tap selector		Categorized into B, C, D & DE four sizes by insulation level							
Mechanical life		1,500,000 operations							
Maintenance-free times		300,000 operations							
Oil compartment of diverter switch	Service pressure	0.03MPa							
	Leakage test	No leakage under 0.1 MPa for 24 hours							
	Over pressure protection	Rupture disc bursts at 0.3 ± 20% MPa							
	Protective relay	Set oil flow speed at 1.0m/s ± 10%							
Motor drive unit		SHM Series or CMA7							

3. Type designation

3.1 Type designation

Due to the different combinations of number of phases, maximum rated through current, the highest voltage for equipment, selector size and connections, type SHZV comes with various models. Hence, the type designation shall provide all the above technical parameter and below is its detailed explanation in Fig. 3 1.

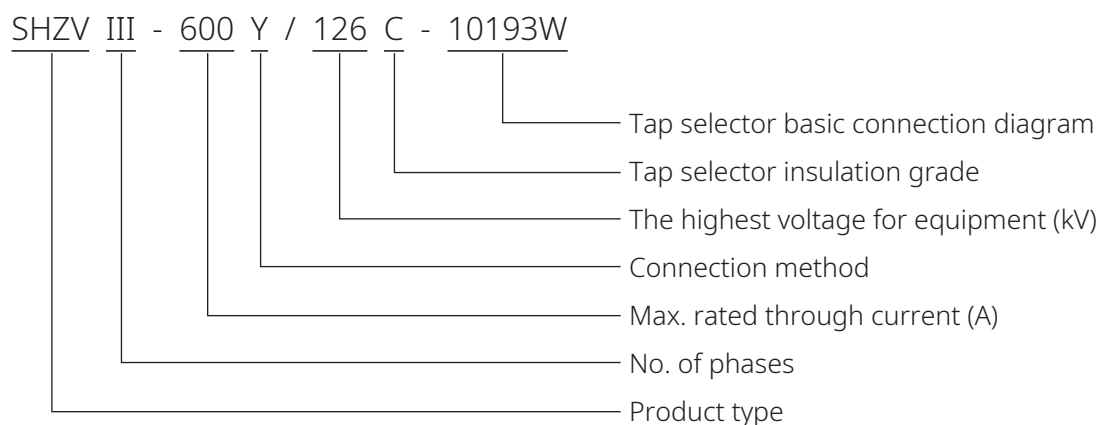


Fig. 3-1 Tap-Changer Type Explanation

3.2 Tap selector basic connection method

Because of voltage regulation range difference and winding connection variations, tap selector has a number of different specifications. Tap selector specification is decided by number of inherent contacts, number of operating positions, number of mid positions and type of change-over selector. Please refer to Fig. 3-2. for indications of different tap selector parameters.

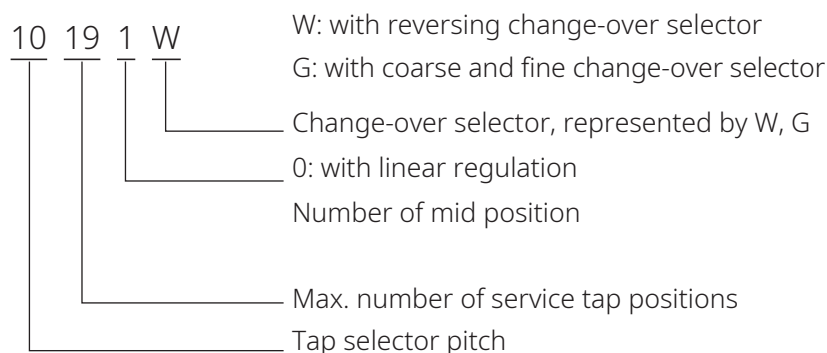


Fig. 3-2 Tap Selector Basic Connection Method Explanation

3.3 Tap selector basic connection diagram

Different transformer tapping corresponds to different tap selector basic connection diagrams. Fig. 3 3 shows common basic connection diagrams. Special requirement can also be specially designed.

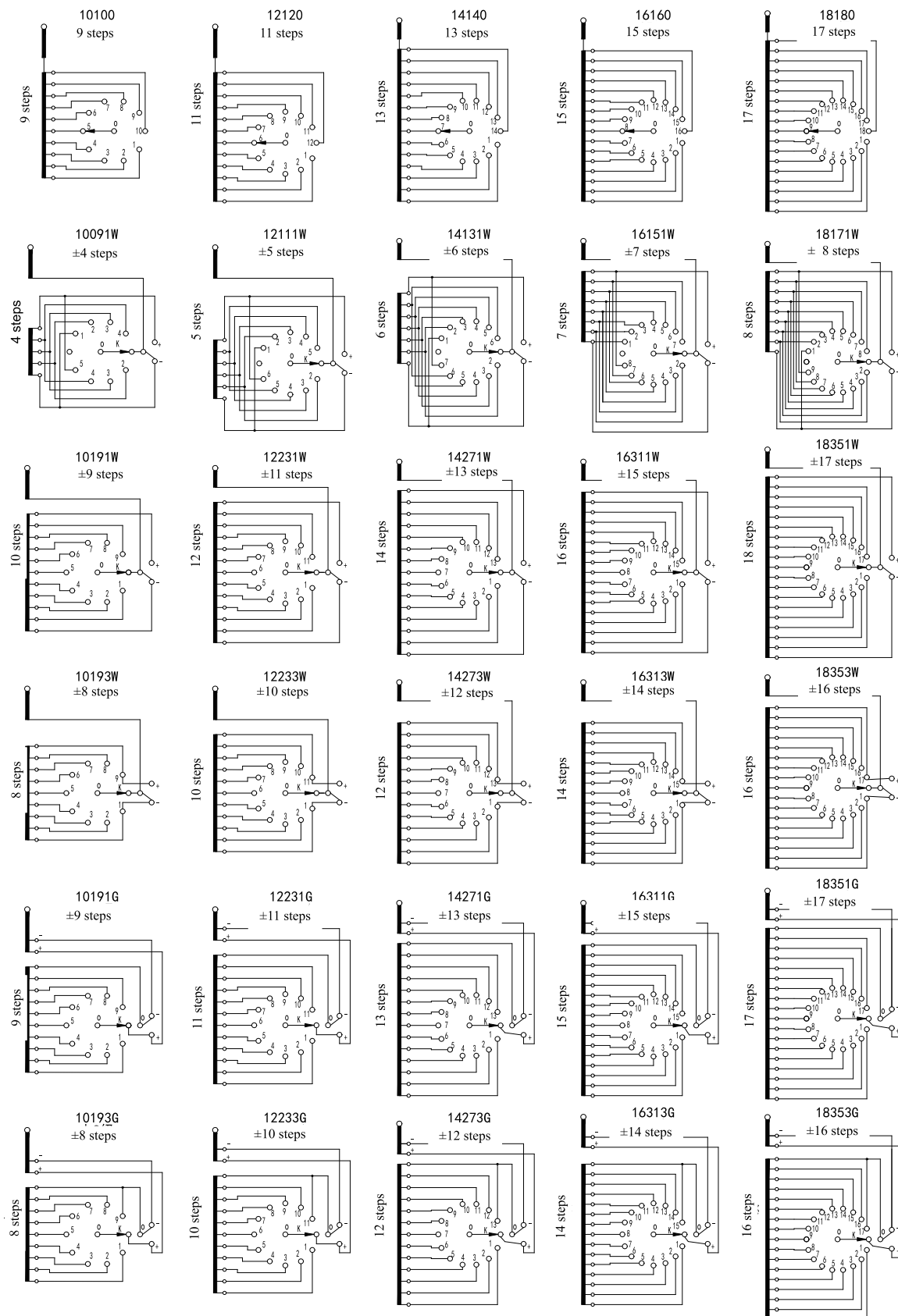


Fig. 3-3 Tap Selector Basic Connection Diagram

4. Terms and definitions

4.1 Rated through-current

Rated through current I_u : The current flowing through an on-load tap-changer toward the external circuit, which can be transferring from one tap to the other at the relevant rated step voltage and which can be carried continuously while meeting the requirement.

Maximum rated through-current I_{um} : The highest rated through-current for which the tap-changer is designed for and which forms the basis for all current related tests.

4.2 Step voltage

Rated step voltage U_i : For each value of rated through current, the highest permissible voltage between terminals which are intended to be connected to successive taps of the transformer.

Maximum rated step voltage U_{im} : The highest value of the rated step voltage for which the tap-changer is designed. The maximum rated step voltage for type SHZV OLTC is 4000V.

4.3 Breaking capacity

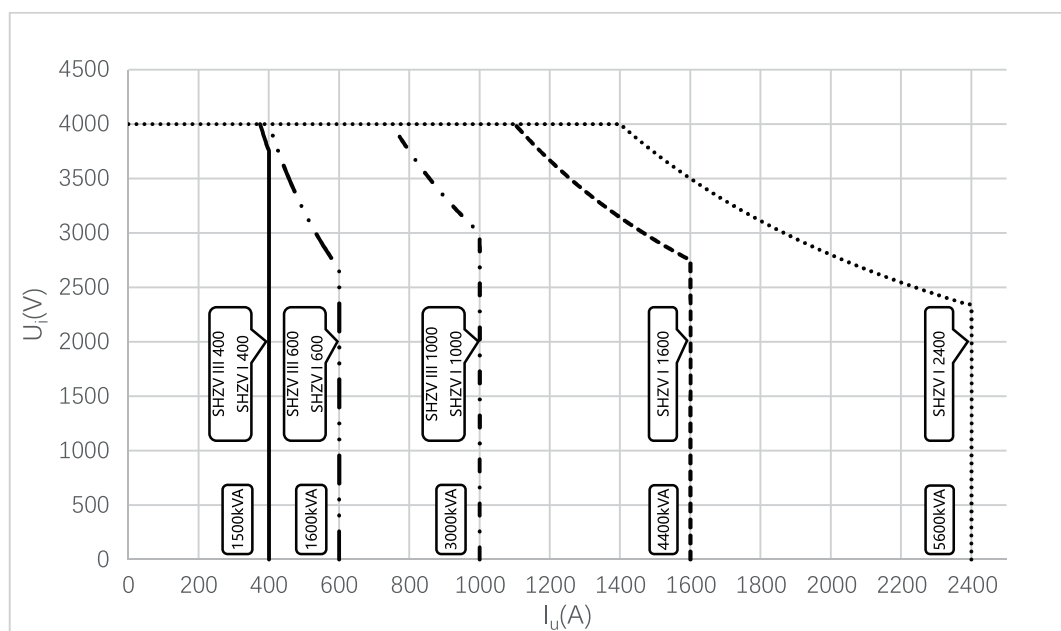


Fig. 4 -1 Curve of SHZV Rated Step Capacity

According to stipulations of IEC 60214-1:2014 and GB/T 10230.1-2019, tap-changer shall be able to break two times of maximum rated through- current and its relevant step voltage for 40 operations. Breaking capacity:

$$P_{st,max}=2P_{StN} \approx 2I_{um} \times U_{StN}$$

P_{StN} : rated step capacity

I_{um} : maximum rated through-current

U_{StN} : relevant rated step voltage

4.4 Electrical life of vacuum interrupter

The electrical life of vacuum interrupt is 600,000 operations

4.5 Short-circuit current test

According to IEC 60214-1:2014 and GB/T 10230.1-2019, all contacts continuously carrying the current shall be able to withstand 2s ($\pm 10\%$) short circuit test current without melting, deformation or mechanical damage. Meanwhile the starting peak current value shall be 2.5 ($\pm 5\%$) times of the root means square value of rated short circuit test current. Refer the short circuit test current values to Table 2-1 of Tap- Changer.

4.6 Service condition of tap-changer

1. Service temperature range of tap-changer in oil is $-25^{\circ}\text{C} \sim +100^{\circ}\text{C}$
2. Service ambient air temperature range of tap-changer is $-25^{\circ}\text{C} \sim +40^{\circ}\text{C}$ and humidity $\leq 85\%$.

Operating ambient air temperature range of tap-changer designed according to the standard is $-25^{\circ}\text{C} \sim +40^{\circ}\text{C}$. If the service ambient temperature is $>40^{\circ}\text{C}$ or $<-25^{\circ}\text{C}$, it should be indicated in the order specifications. The material and accessories will be specially treated to suit the needs.

3. Perpendicular deflection between ground and tap-changer after being mounting on transformer shall be less than 2%.
4. There shall be no serious dust, explosive gas or corrosive gas on service site.

Remark: For more severe environmental conditions of tap changers or motor drive units, users need to contact the technical department of Huaming for confirmation.

4.7 Tap-changer insulation level to ground

Tap-changer insulation level to earth is the insulation between tap-changer live parts and grounding part. It is determined by dielectric tests according to IEC 60214-1:2014 and GB/T 10230.1-2019 (Please refer to Table 4 1) The requirement of which correlates to the transformer tap winding location, regulation range & regulation method, winding connection & arrangement and rated voltage of transformer winding. It's decided by the insulation to earth of transformer tap winding.

Table 4 -1 Tap- Changer Insulation Level to ground

(unit: kV)

The highest voltage for equipment U_m	Rated separate source AC withstand voltage (50Hz, 1min)	Rated lightning impulse withstand voltage (1.2/50 μ s)
72.5	140	350
126	230	550
170	325	750
252	460	1050
300	480	1100
363	510	1175

4.8 Insulation Classification and Internal Withstand Voltage Levels

The internal insulation level of the on-load tap-changer (OLTC) is classified into four categories: B, C, D, and DE, in increasing order of dielectric strength. The power frequency and lightning impulse withstand voltages vary depending on the insulation class, for:

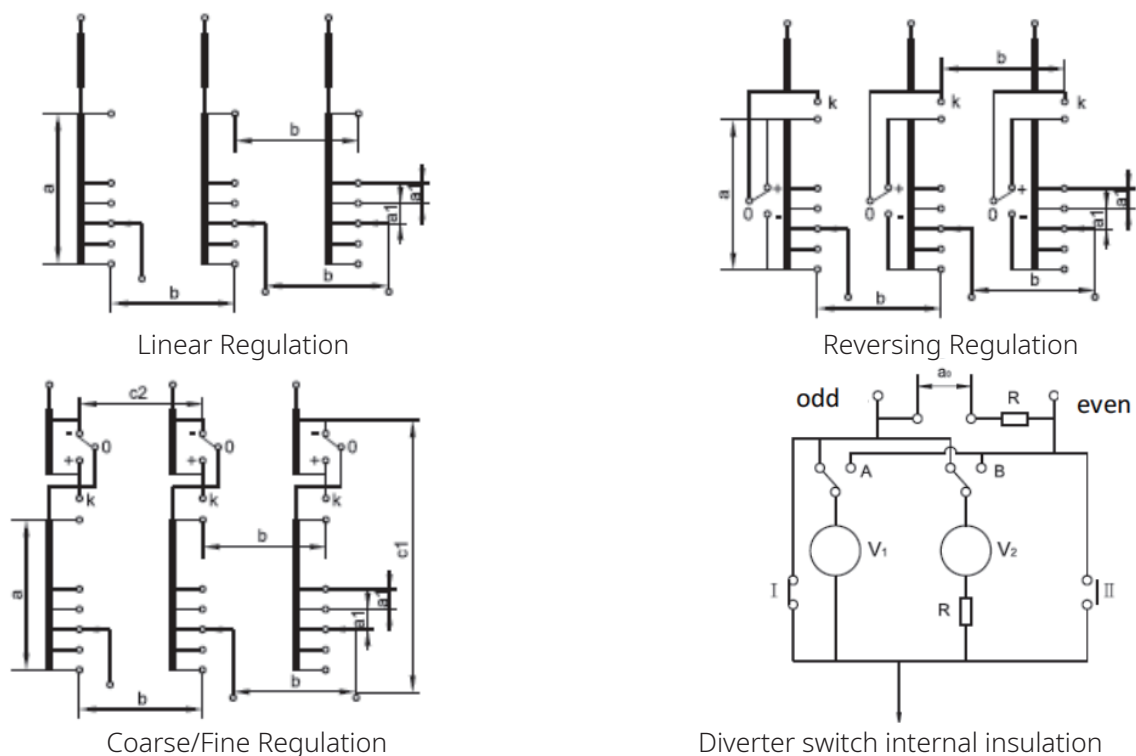
- Step-to-step insulation,
- Highest-to-lowest tap insulation,
- Phase-to-phase insulation.

Class B has the lowest withstand voltage ratings, while class DE has the highest.

Potential Distribution and Dielectric Requirements

Figure 4.2 illustrates the insulation distance designation across the OLTC when connected to the regulating winding. Table 4.2 specifies the power frequency (50 Hz, 1 min) and lightning impulse (1.2/50 μ s) withstand voltages for each insulation distance.

Fig. 4.2 – OLTC Basic Connection Diagram and Insulation Distance Designation



Definitions of Insulation Distances:

- a: Between start and end of a fine tap winding; or between start and end of a coarse tap winding.
- b: Between any tap of different fine tap windings, or between ends of different coarse windings.
- a_0 : Between selected and preset diverter switch taps.
- a_1 : Between any selected and preselected taps of the tap selector.
- c_1 : Between the start of the coarse tap winding and the current take-off terminal (same phase).
- c_2 : Between start contacts (–) of coarse windings (different phases).

Key OLTC Components:

- T_1, T_2 : Isolating contacts.
- V_1, V_2 : Vacuum interrupters.
- R_1, R_2 : Transition resistors.
- A: Odd-numbered shunt contact.
- B: Even-numbered shunt contact.

Table 4-2 Tap-Changer Internal Insulation Level

(unit: kV)

Insulation distance mark	Tap selector size B		Tap selector size C		Tap selector size D		Tap selector size DE	
	1.2/50 μ s	50Hz, 1min	1.2/50 μ s	50Hz, 1min	1.2/50 μ s	50Hz, 1min	1.2/50 μ s	50Hz, 1min
a	265	50	365	82	460	105	550	120
b	265	50	350	82	460	146	550	160
a_{01}	90	20	90	20	90	20	90	20
a_{02}	70	20	70	20	70	20	70	20
a_1	150	30	150	30	150	30	150	30
c_1	500	145	550	180	590	225	660	230
c_2	500	145	550	195	590	225	660	250

Notes:

- a_{01} (Spark gap protection): Lightning impulse (1.2/50 μ s) withstand range: **90–130 kV** (100% response at **130 kV**).
- a_{02} (ZnO arrester protection): Lightning impulse (1.2/50 μ s) withstand range: **50–70 kV** (100% response at **70 kV**).

4.9 Tap-changer mounting method

Type SHZV tap-changer is mounted to transformer tank top by a head flange. Hence, a mounting flange shall be provided by transformer producer, the dimension of which shall refer to the drawing of Appendix 36. Type SHZV tap-changer is only applicable to bell type mounting. The supporting flange of the tap-changer is only for temporary support during the transformer conductor connection. After putting the bell tank, tap-changer shall be fixed to the mounting flange of the transformer. The connection pipes on head flange of type SHZV tap-changer is of spread around arrangement (Refer details to appendix 33, appendix 34).

5. Special designs

5.1 Potential connection of the tap winding

For transformers with high voltage rating and big regulation range, during the operation of the change-over selector, the tap winding is disconnected momentarily from the main winding and in a so-called "suspension" status. At that moment, the tap winding takes a new potential which is determined together by the coupling capacitance to ground C_e and coupling capacitance to the adjacent winding C_w (refer details to Fig. 5 2). Usually this potential is different from the previous potential of the tap winding before the operation. The difference between the two is called bias voltage. This bias voltage turns out to be the recovery voltage on the gap of the change-over selector. When the recovery voltage exceeds a certain critical value, the change-over selector would discharge electricity and produce considerable amount of gas. This could be a serious problem. Therefore, potential connection of the tap winding must be considered when this recovery voltage exceeds a certain value, in order to avoid the discharge during the operation of the change-over selector.

The permissible recovery voltage to type SHZV tap-changer is 35kV. In case the recovery voltage of the change-over selector exceeds this value, a tie-in resistor with fixed value shall be permanently connected into the tap winding (refer to Fig. 5 1). The mounting location and dimension of tie-in resistor for SHZV can be found in Appendix 40.

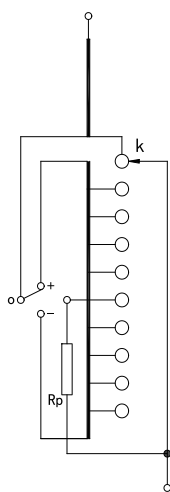


Fig. 5-1 Potential Connection by Tie-In Resistor

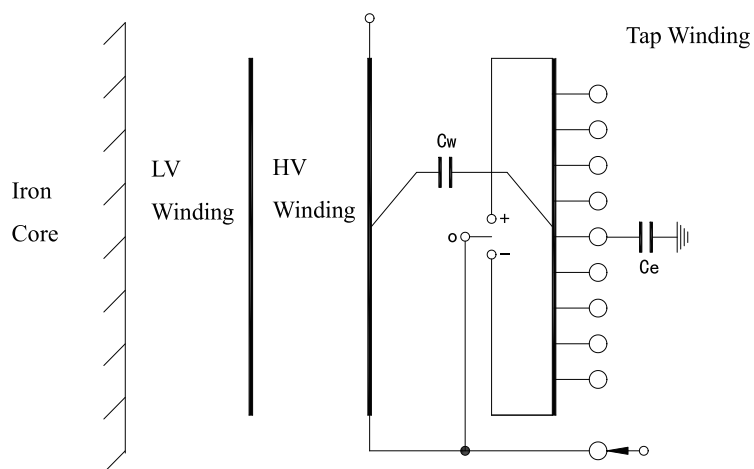


Fig. 5-2 Arrangement of Winding and Coupling Capacitance

For the convenience of Huaming to define the load of the change-over selector and size & quantity of the tie-in resistor, please provide the following transformer data when selecting tap-changer:

1. Complete transformer data: rated capacity, rated voltage, regulation range, connection of winding and insulation level, etc.
2. Arrangement of winding: relative location between tap winding and adjacent winding or winding part
3. Operating A.C. voltage across windings or layers of windings adjacent to the tap windings
4. Capacitance between tap winding and adjacent winding (C_w)
5. Capacitance of the tap winding to ground or grounded adjacent windings (if exist) (C_e)
6. Voltage stress across half the tap winding at lightning impulse voltage test
7. A.C. voltage across half the tap winding under operation and test conditions.

5.2 Two phase and single phase of type SHZV

Type SHZV tap-changer can be designed as one motor drive unit (or three motor drive unit) driving three single phase tap changers or one two phases plus one single phase tap-changer, for regulation of delta connection or other regulation locations other than neutral point.

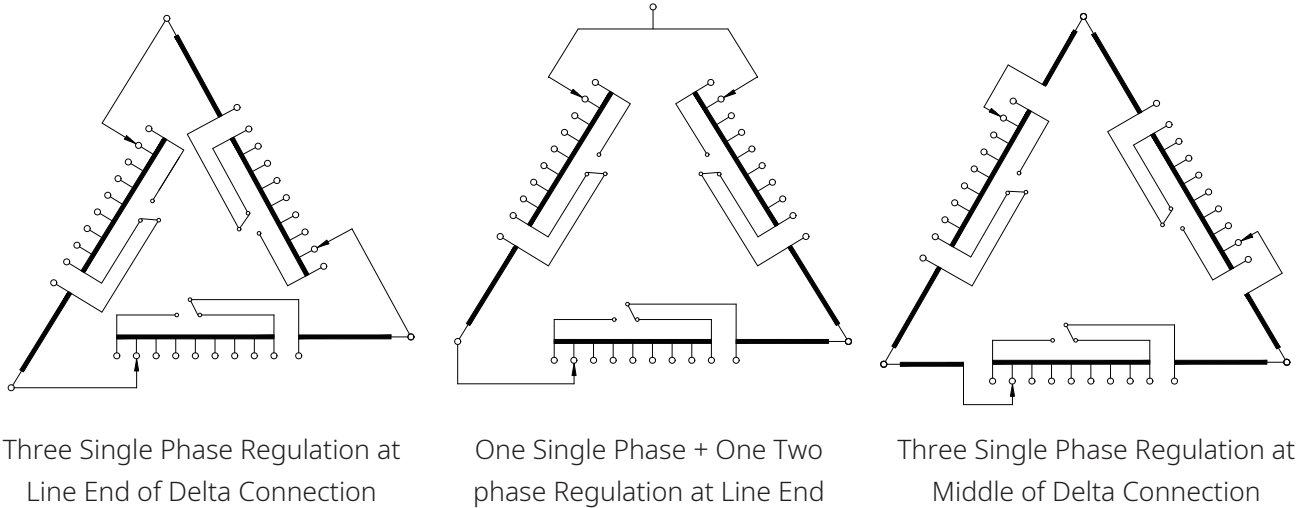


Fig. 5 -3 Basic Connections for Delta-connected Transformer Winding

5.3 Structures and application

Common structure: It is suitable for the contact pitch of 10, 12 and 14 in linear, reversing and coarse and fine regulating.

Strip-type selector structure: SHZV strip-type selector is the same as the tap selector of type CM. Thus it can meet the technical data of CM tap selector.

Note: The maximum current of SHZV with cage tap selector is 600A for 3-phase and 1500A for single phase. Please refer to CM technical data for connection diagram or contact us.

Please contact us when choosing SHZV with strip-type selector for the drawings (in linear, reversing regulation or in other regulation) which are excluded in this SHZV technical data.

5.4 Step protection and application

SHZV over voltage between steps a0 protection can be realized via spark gaps or varistor (ZnO).

The spark gap protection mechanism has a simple structure and small size. When the over voltage attacks, the breakdown first happens to the spark gap so as to protect the insulation between steps. After that, the spark gap must be replaced to ensure the tap-changer's normal operation.

The varistor (ZnO) has the same function and better performance. The size, however, is bigger. So usually the spark gap is adopted. If the varistor (ZnO) is needed, please specify in the order specification.

6. Motor drive unit

OLTC may operate by motor drive unit according to the requirement, please refer to table 6 1 for technical data.

Table 6-1 Technical Data of Motor Drive Unit

Motor drive unit		SHM-D	CMA7	
Item				
Motor	Rated power (W)	750	750	1100
	Rated voltage (V)	200~240 (AC)	3-Phase 380 (AC)	
	Rated current (A)	4	2.0	2.8
	Rate frequency(Hz)	50 or 60	50 or 60	
	Rotate speed (r/min)	330 direct drive	1400	
Rated torque on drive shaft (Nm)		35 static torque	18	26
Revolution of the drive shaft per switching operation (turns)		33	33	
Revolution of the hand crank per switching operation (turns)		33	33	
Running time per switching operation (s)		about 6	about 5	
Max. operation positions		107	107	
Voltage for control circuit and heater circuit (V)		220 (AC)	220 (AC)	
Heater power consumption (W)		100	50	
A.C. voltage test to ground (kV/50Hz, 1min)		2	2	
Approx. weight (kg)		80	90	
Protective degree		IP66	IP66	
Mechanical life (operations)		≥2,000,000	≥800,000	

7. Operation controllers

7.1 SHM-KX remote digital controller

SHM-KX remote digital controller (hereinafter referred to as SHM-KX) can be used together with motor to realize manual and automatic control of tap-changer. SHM-KX uses LCD screen, which is connected with the local control module for bilateral digital communication.

SHM-KX main technical data is as below:

Working voltage: AC 85V to 264V or DC 100-250V
Power consumption: about 7W
Environment temperature: 0°C to 50°C Indoor; Comparative humidity ≤ H95% (no condensation)
Installation Method: Panel or cabinet mounting.

7.2 HMC-3C position indicator

HMC-3C OLTC position indicator is a support fitting for motor drive unit, it can be used to indicate the OLTC position, and has the function of "1-> N", "STOP", "N-> 1" control as well as remote control indicator lamp, its input is decimal code and output is BCD code.

HMC-3C Monitor Technical Parameters:

Operating Voltage: 220 V (AC)	Operating Frequency: 50 Hz
Maximum Number of Display position Number: 107	Operating Temperature: -10 °C to +40 °C

Note: Specify any other power supply technical requirements for the controller at the time of order.

8. Accessories

8.1 Bevel gearbox

Bevel gearbox is used for the inter-connection of tap-changer horizontal shaft and vertical shaft, in order to transfer the driving torque from motor drive unit to the tap-changer. Please refer to Appendix 41 for the overall dimension.

8.2 Protective relay

Protective relay is the one of protective devices for oil-immersed on-load tap-changer, when OLTC interior failure produces gas and oil surge, the protective relay contact acts, and switches on to the tripping circuit of the transformer circuit breaker, the transformer will be cut off at once.

Protective relay is mounted onto the connection pipe between OLTC head and oil conservator; make sure that protective relay marked with arrowhead side shall be connected to oil conservator. There are multiple types of protective relays configured for tap-changer of Huaming for users to select. See Appendix 47 for details of installation dimensions of protective relays. (If the user has any special requirements, please contact the technical department of our company)

8.3 Pressure relief valve

Pressure relief valve and rupture disc are safety protection devices of the on-load tap-changer. In case tap-changer has an internal failure, which decomposes the oil in the compartment and produces large amount gas, the internal pressure inside the oil compartment will increase dramatically. If this pressure couldn't be released, tap-changer will deform or even explode. Therefore, pressure relief devices can avoid the upgrade of the failure.

Pressure relief valve is a self-sealing relief valve. It opens the cover in case of over pressure and re-closes after the pressure is released, which can be used repeatedly and minimize the liquid loss during the operation.

The rupture disc is a weak point on the top cover of tap-changer. Once the pressure in the oil compartment exceeds the set value, the disc will explode to release the over pressure of the compartment, as a result the oil compartment will be prevented from damage.

Pressure relief valve is a low-energy failure protection device. The rupture disc is a high-energy protection device. Tap-changer failure usually tends to be high-energy failure. Hence, rupture disc is recommended for tap-changer, or use it as an assistant protection besides the rupture disc. Pressure relief valve is an optional accessory of tap-changer for customer to select when ordering the tap-changer.

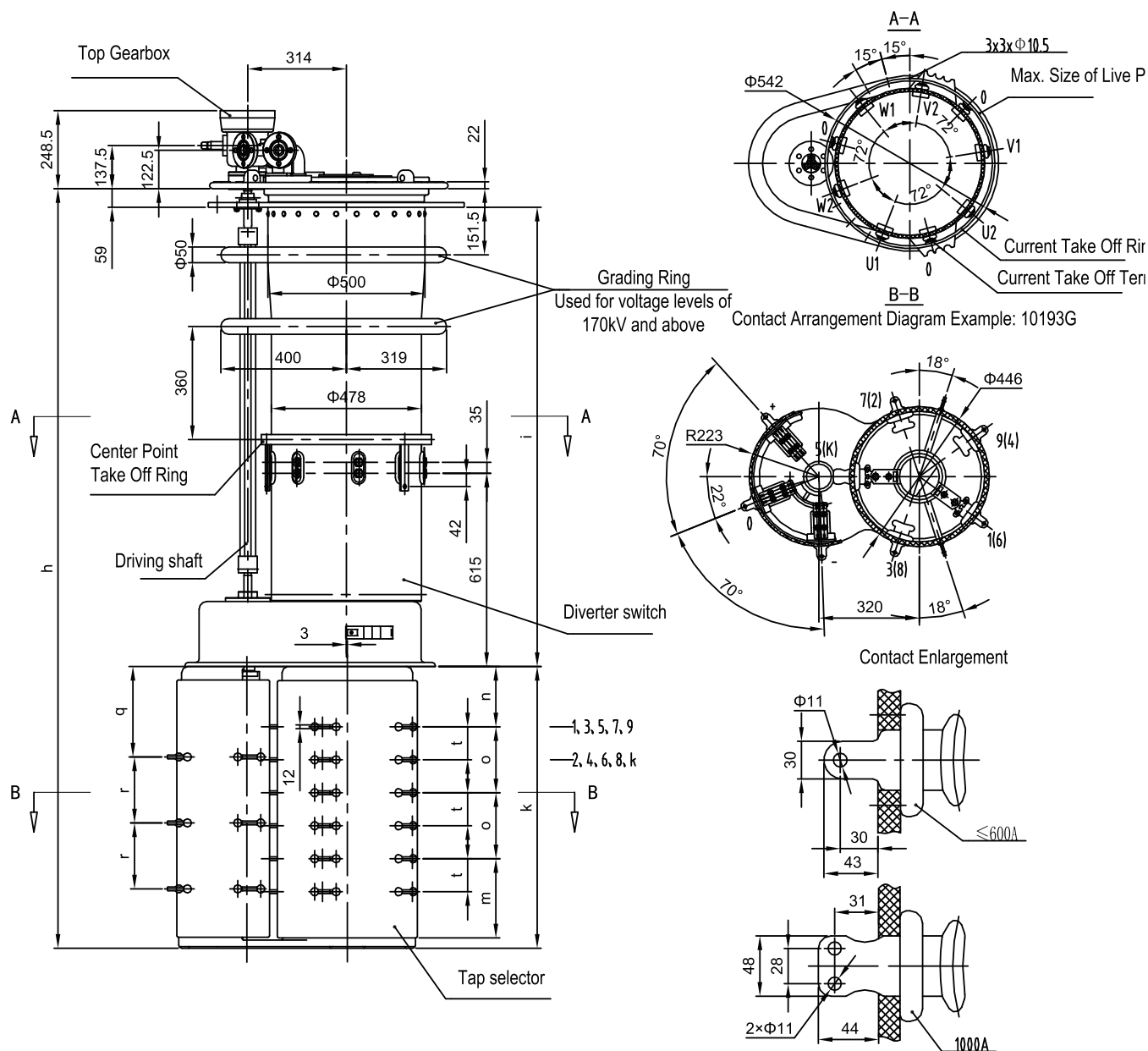
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Unit: mm

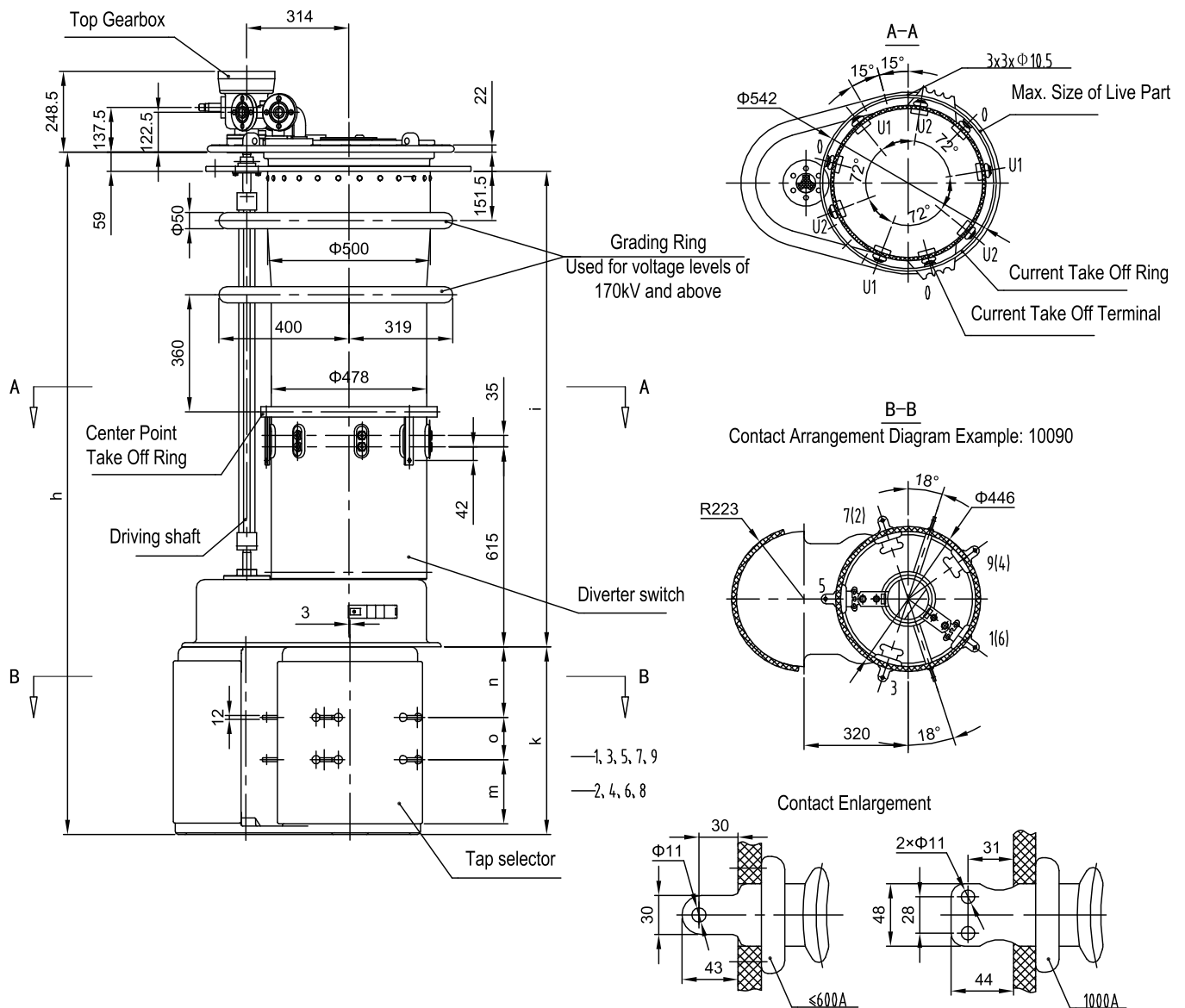
Appendix 3 SHZV III 400/600/1000A overall dimensions with coarse change-over selector, cylinder tap selector



Switch model	SHZV III 400/600/1000/72.5~363/B						SHZV III 400/600/1000/72.5~363/C						SHZV III 400/600/1000/72.5~363/D					
The highest voltage for equipment (kV)	72.5	126	170	252	300	363	72.5	126	170	252	300	363	72.5	126	170	252	300	363
Size (mm)	h	2173	2303	2433	2533	2663	2348	2478	2608	2708	2758	2838	2818	2948	3078	3178	3228	3308
	i	1217	1347	1477	1577	1707	1217	1347	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	897					1072					1542						
	n	192					217					282						
	o	210					260					390						
	t	105					130					195						
	m	252					302					432						
	q	288					338					468						
	r	210					260					390						
Oil volume (dm ³)	130	150	170	190	200	210	130	150	170	190	200	210	130	150	170	190	200	210
Oil draining (dm ³)	199	224	244	264	274	284	200	225	245	265	275	285	207	232	252	272	282	292
Weight (kg)	280						285						290					

Unit: mm

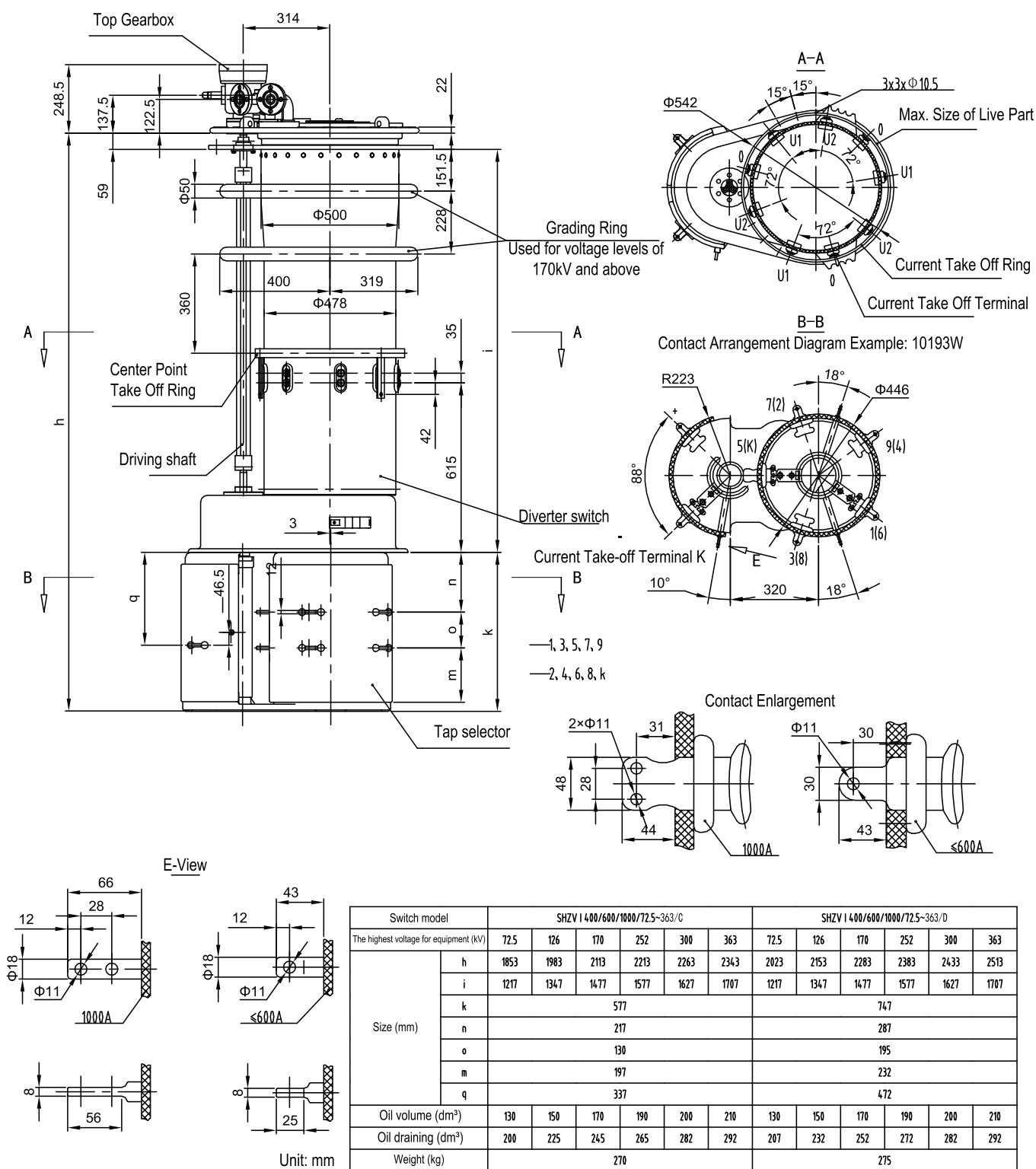
Appendix 4 SHZV I 400/600/1000A overall dimensions without change-over selector, cylinder tap selector



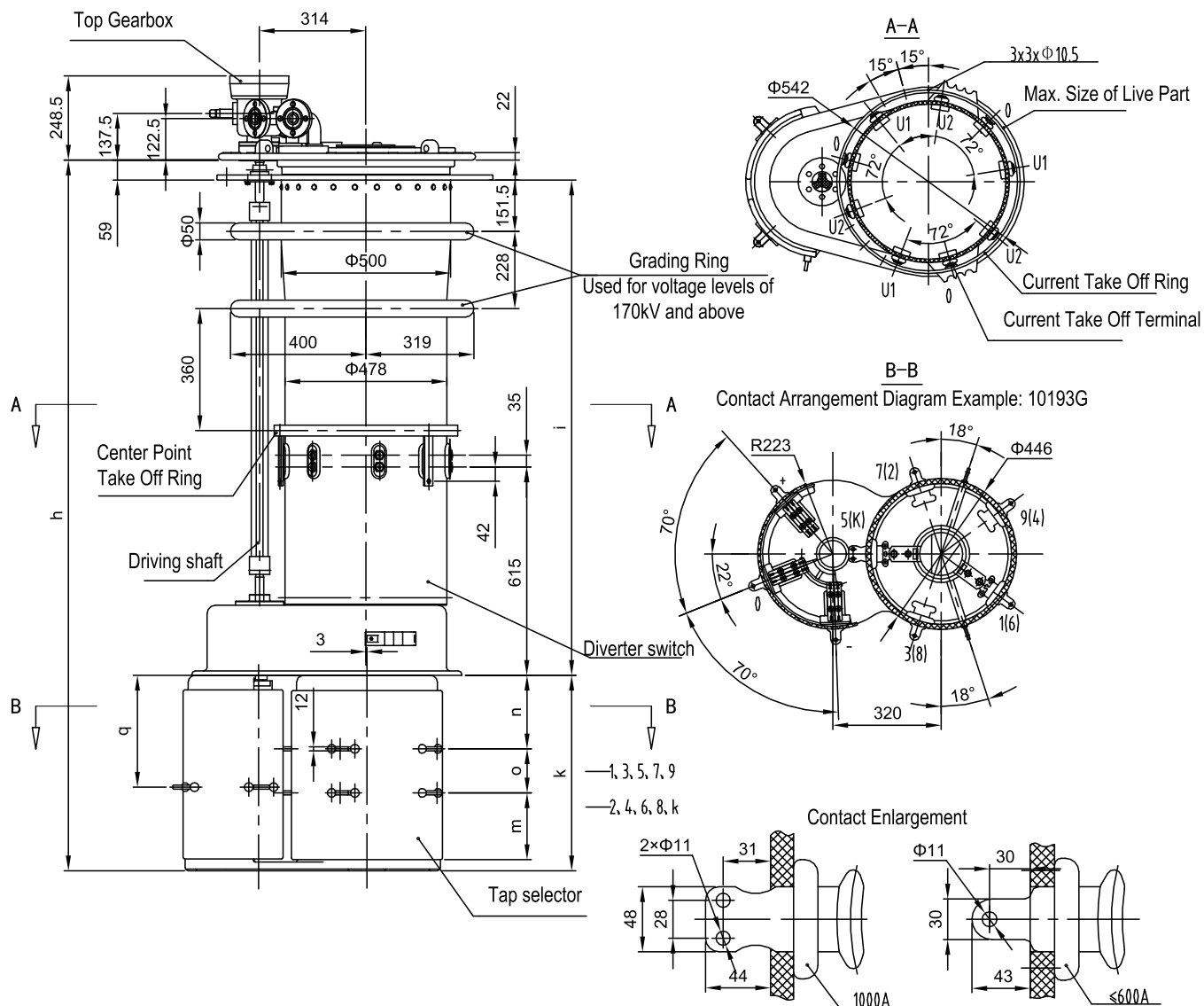
Unit: mm

Switch model		SHZV I 400/600/1000/72.5~363/C						SHZV I 400/600/1000/72.5~363/D					
The highest voltage for equipment (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Size (mm)	h	1853	1983	2113	2213	2263	2343	2023	2153	2283	2383	2433	2513
	i	1217	1345	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	577						747					
	n	217						287					
	o	130						195					
	m	197						232					
Oil volume (dm³)		130	150	170	190	200	210	130	150	170	190	200	210
Oil draining (dm³)		200	225	245	265	282	292	207	232	252	272	282	292
Weight (kg)		270						275					

Appendix 5 SHZV I 400/600/1000A overall dimensions with reversing switch, cylinder tap selector



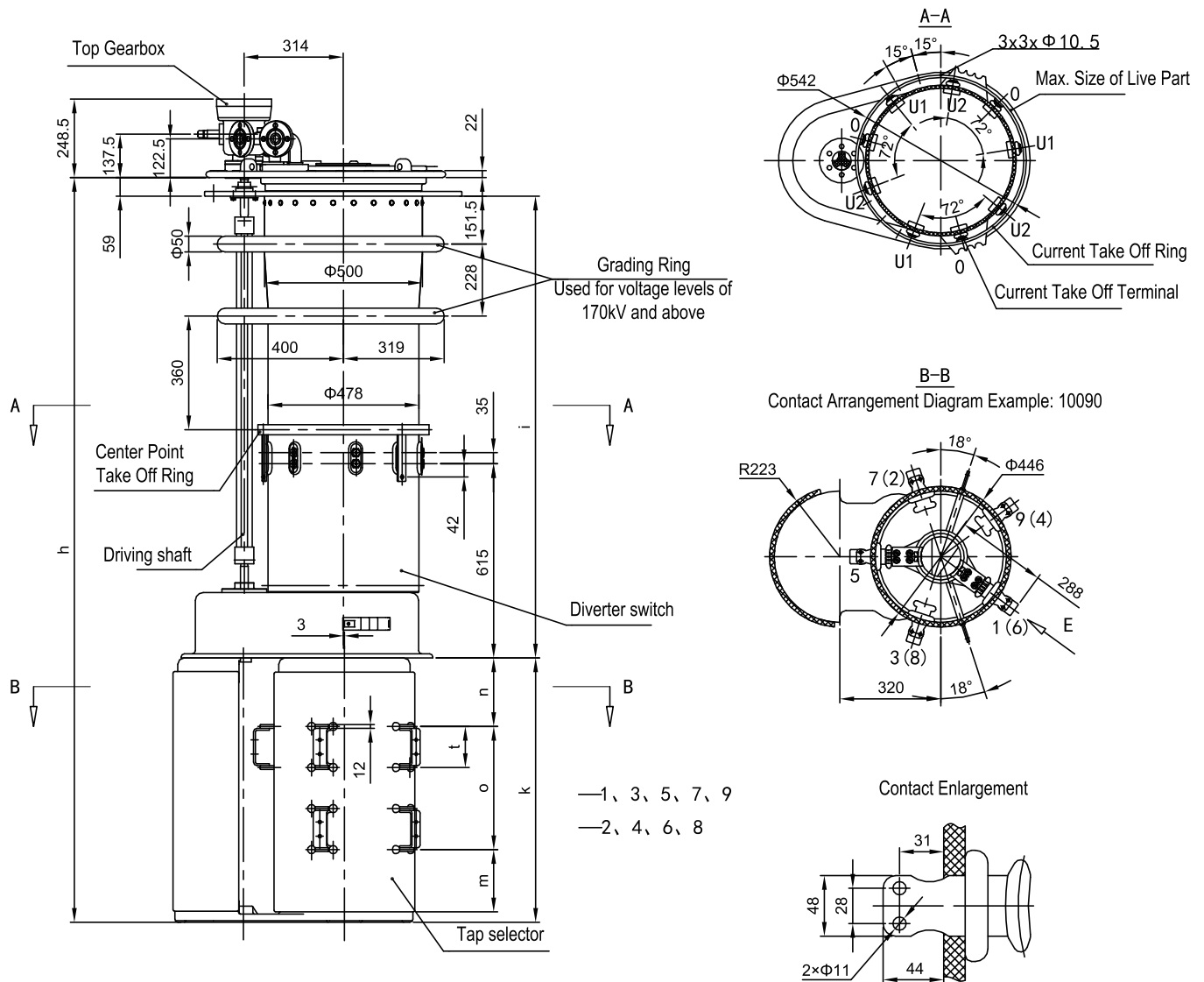
Appendix 6 SHZV I 400/600/1000A overall dimensions with coarse change-over selector, cylinder tap selector



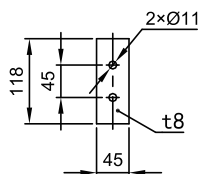
Switch model		SHZV I 400/600/1000/72.5-363/C						SHZV I 400/600/1000/72.5-363/D					
The highest voltage for equipment (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Size (mm)	h	1853	1983	2113	2213	2263	2343	2023	2153	2283	2383	2433	2513
	i	1217	1347	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	577						747					
	n	217						287					
	o	130						195					
	m	197						232					
	q	330						465					
Oil volume (dm ³)		130	150	170	190	200	210	130	150	170	190	200	210
Oil draining (dm ³)		200	225	245	265	282	292	207	232	252	272	282	292
Weight (kg)		270						275					

Unit: mm

Appendix 7 SHZV I 1600A overall dimensions without change-over selector, cylinder type



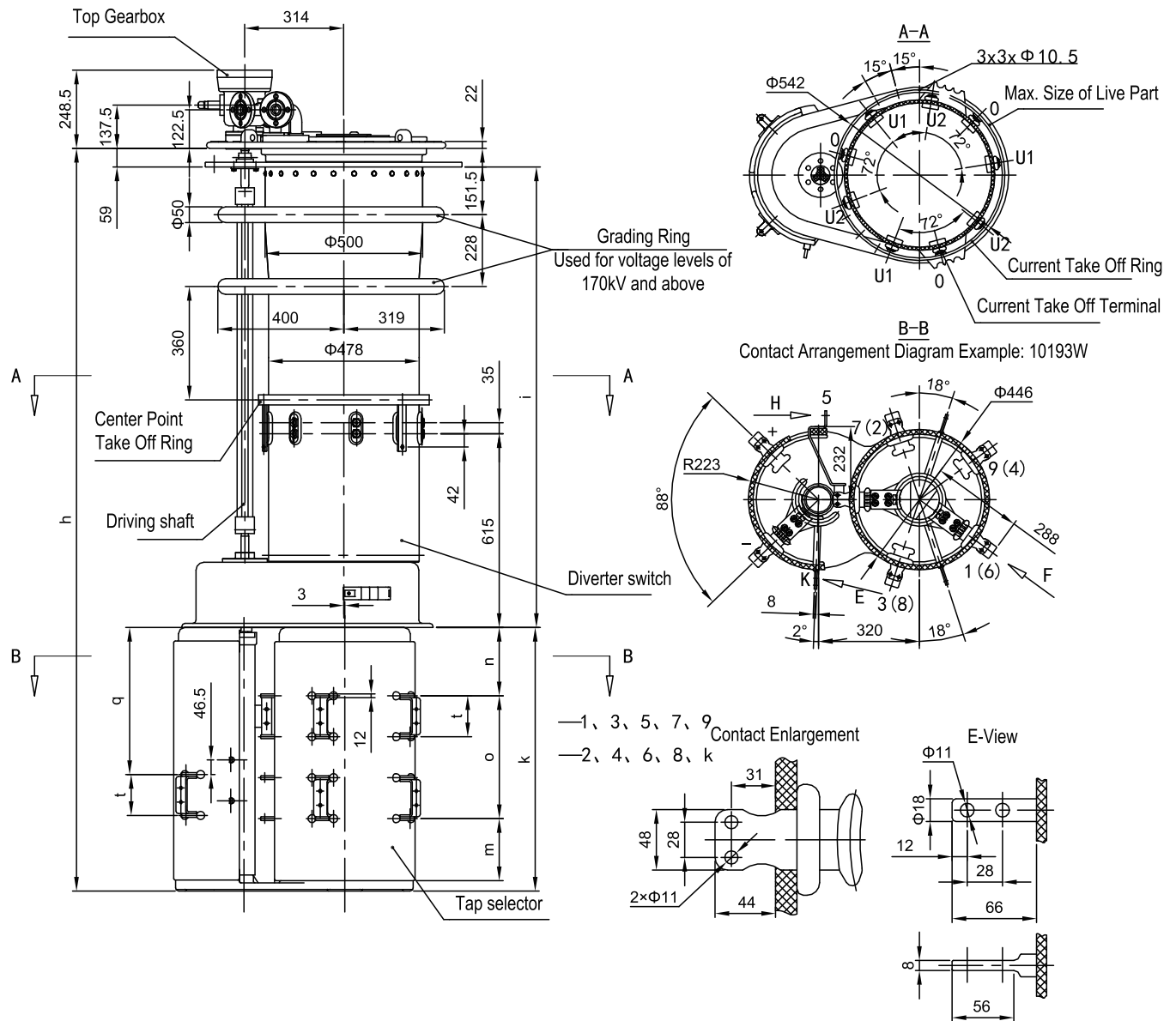
E-View
Rotation



Unit: mm

Switch model		SHZV I 400/600/1000/72.5~363/C						SHZV I 400/600/1000/72.5~363/D					
The highest voltage for equipment (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Size (mm)	h	1853	1983	2113	2213	2263	2343	2023	2153	2283	2383	2433	2513
	i	1217	1345	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	577						747					
	n	217						287					
	o	130						195					
	m	197						232					
Oil volume (dm ³)		130	150	170	190	200	210	130	150	170	190	200	210
Oil draining (dm ³)		200	225	245	265	282	292	207	232	252	272	282	292
Weight (kg)		270						275					

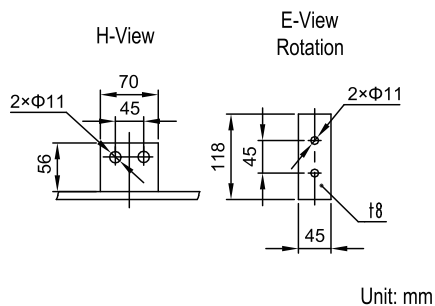
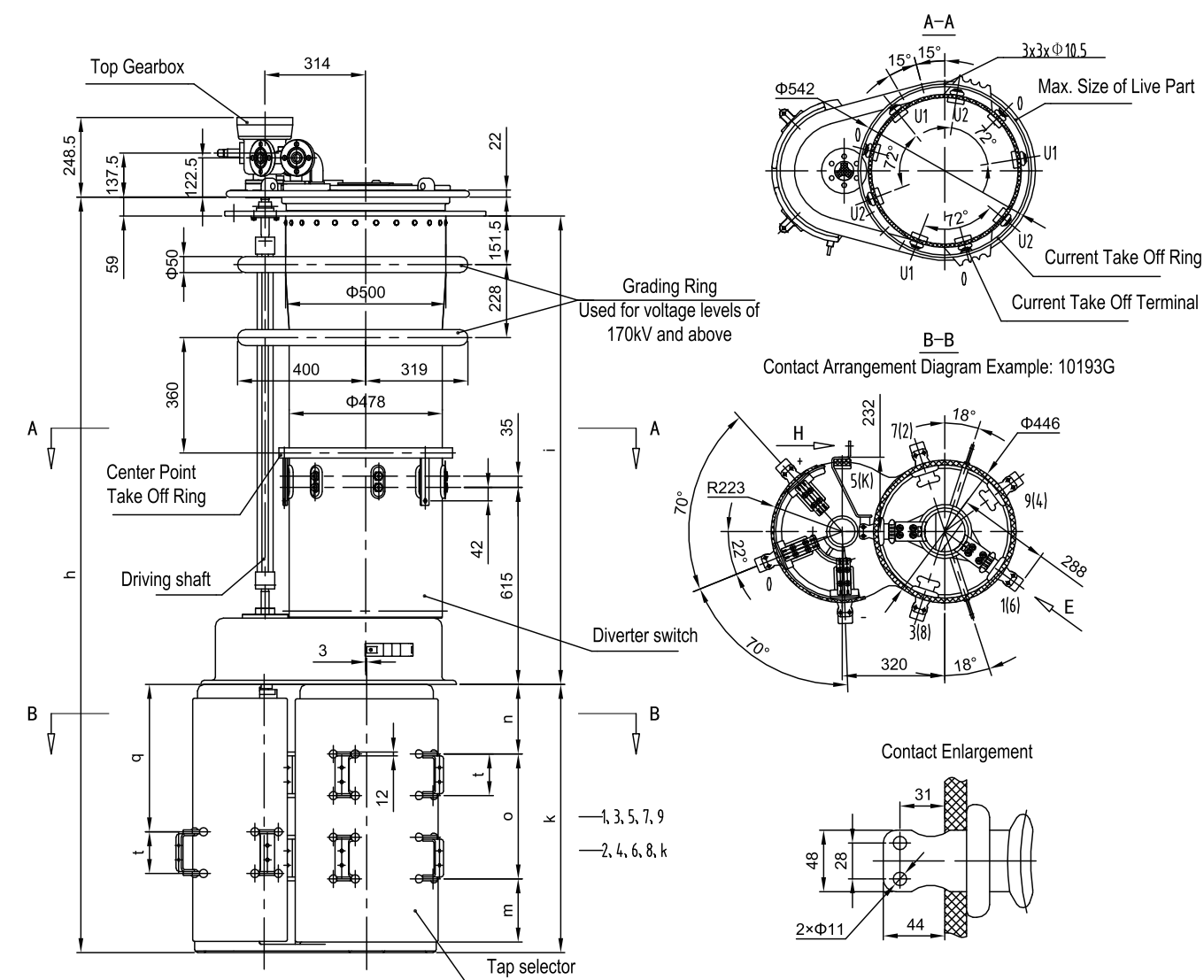
Appendix 8 SHZV I 1600A overall dimensions with reversing switch, cylinder tap selector



Switch model		SHZV I 1600/72. 5~363/C						SHZV I 1600/72. 5~363/D					
The highest voltage for equipment (kV)		72. 5	126	170	252	300	363	72. 5	126	170	252	300	363
Size (mm)	h	2113	2243	2373	2473	2523	2403	2283	2413	2543	2643	2693	2773
	i	1217	1347	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	837						1007					
	n	217						287					
	o	260						325					
	t	130						130					
	m	327						362					
	q	467						602					
Oil volume (dm³)		130	150	170	190	200	210	130	150	170	190	200	210
Oil draining (dm³)		200	225	245	265	275	285	207	232	252	272	282	292
Weight (kg)		280						285					

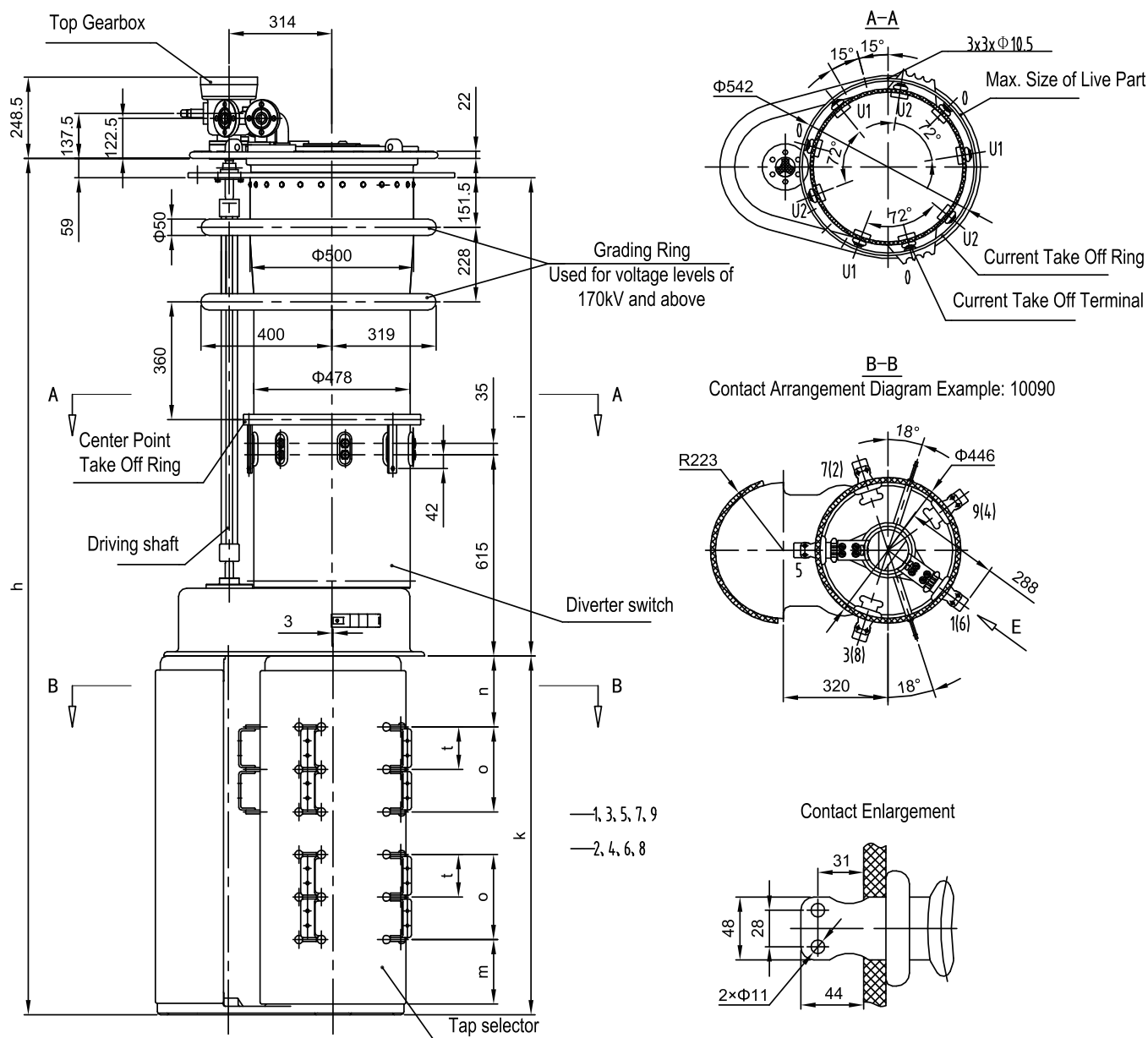
Unit: mm

Appendix 9 SHZV I 1600A overall dimensions with coarse and fine change-over selector, cylinder tap selector

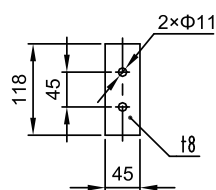


Switch model		SHZV I 1600/72.5~363/C						SHZV I 1600/72.5~363/D					
The highest voltage for equipment (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Size (mm)	h	2098	2228	2358	2458	2508	2388	2268	2398	2528	2628	2678	2758
	i	1217	1347	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	837						1007					
	n	217						287					
	o	260						325					
	t	130						130					
	m	327						362					
	q	460						595					
Oil volume (dm³)		130	150	170	190	200	210	130	150	170	190	200	210
Oil draining (dm³)		200	225	245	265	275	285	207	232	252	272	282	292
Weight (kg)		280						285					

Appendix 10 SHZV I 2400A overall dimensions without change-over selector, cylinder type



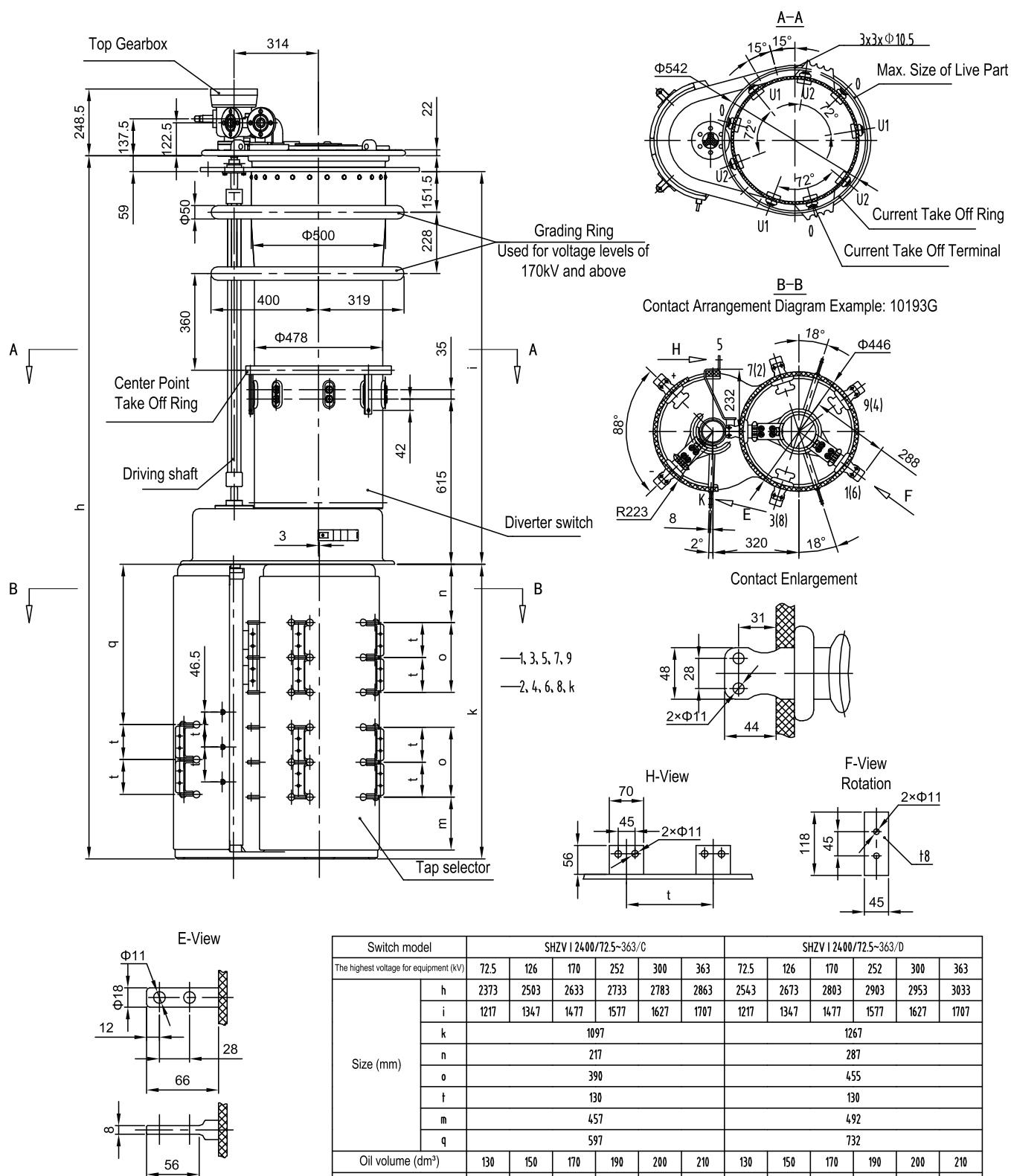
E-View
Rotation



Unit: mm

Switch model		SHZV I 2400/72.5~363/C						SHZV I 2400/72.5~363/D					
The highest voltage for equipment (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Size (mm)	h	2373	2503	2633	2733	2783	2863	2543	2673	2803	2903	2953	3033
	i	1217	1347	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	1097						1267					
	n	217						287					
	o	390						455					
	t	130						130					
	m	457						492					
Oil volume (dm³)		130	150	170	190	200	210	130	150	170	190	200	210
Oil draining (dm³)		200	225	245	265	275	285	207	232	252	272	282	292
Weight (kg)		290						295					

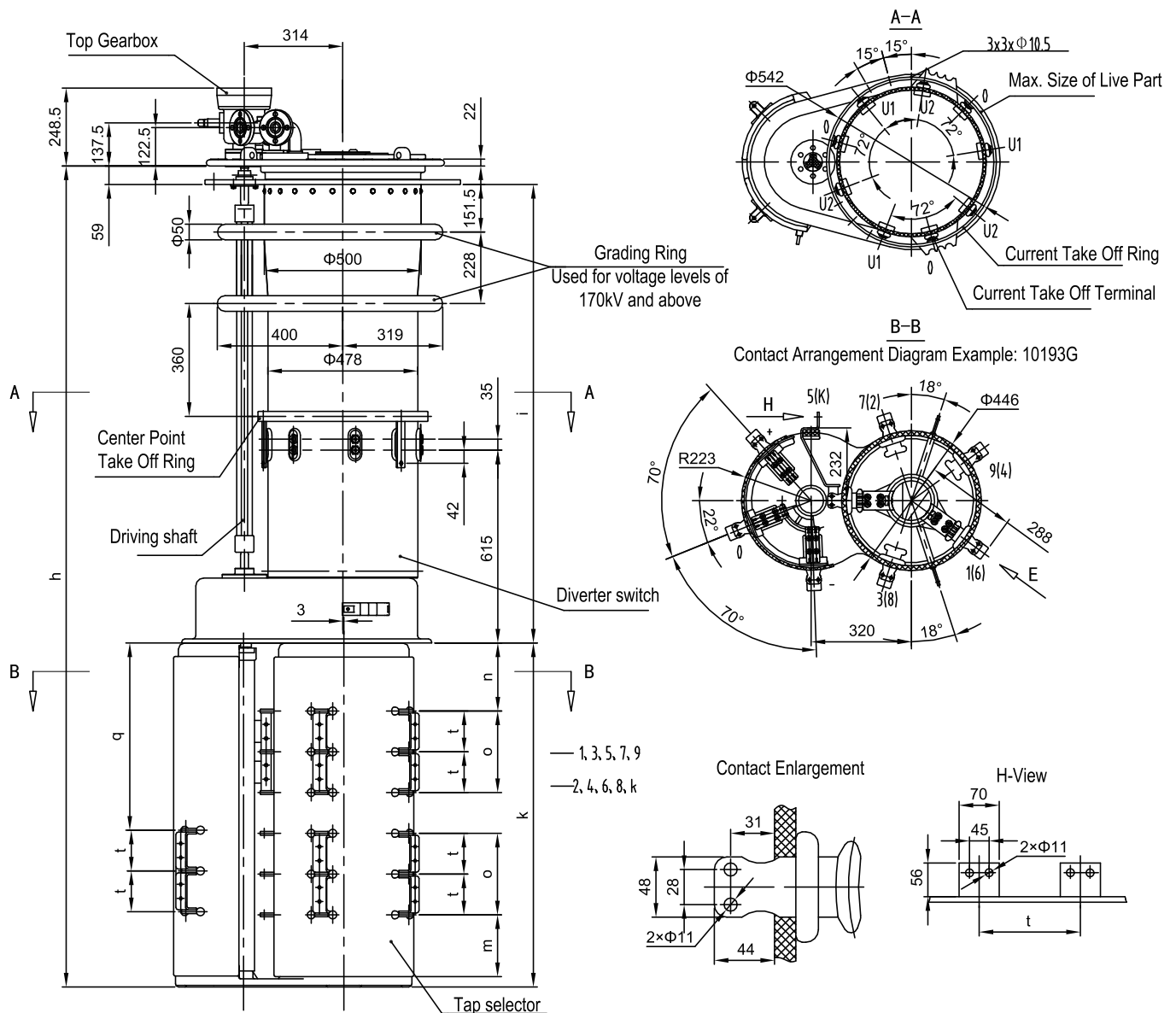
Appendix 11 SHZV I 2400A overall dimensions with reversing switch, cylinder tap selector



Switch model		SHZV I 2400/72.5~363/C						SHZV I 2400/72.5~363/D					
The highest voltage for equipment (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Size (mm)	h	2373	2503	2633	2733	2783	2863	2543	2673	2803	2903	2953	3033
	i	1217	1347	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	1097						1267					
	n	217						287					
	o	390						455					
	t	130						130					
	m	457						492					
	q	597						732					
Oil volume (dm³)	130	150	170	190	200	210	130	150	170	190	200	210	
Oil draining (dm³)	200	225	245	265	275	285	207	232	252	272	282	292	
Weight (kg)	290						295						

Unit: mm

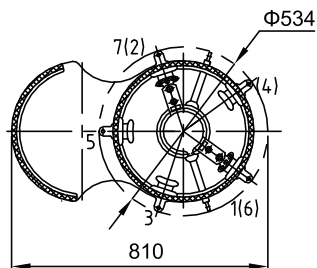
Appendix 12 SHZV I 2400A overall dimensions with coarse and fine change-over selector, cylinder tap selector



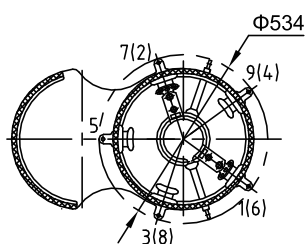
Switch model		SHZV I 2400/72.5-363/C						SHZV I 2400/72.5-363/D					
The highest voltage for equipment (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Size (mm)	h	2373	2503	2633	2733	2783	2863	2543	2673	2803	2903	2953	3033
	i	1217	1347	1477	1577	1627	1707	1217	1347	1477	1577	1627	1707
	k	1097						1267					
	n	217						287					
	o	390						455					
	t	130						130					
	m	457						492					
	q	597						725					
Oil volume (dm³)		130	150	170	190	200	210	130	150	170	190	200	210
Oil draining (dm³)		200	225	245	265	275	285	207	232	252	272	282	292
Weight (kg)		290						295					

Appendix 13 SHZV tap selector contacts arrangement, cylinder tap selector

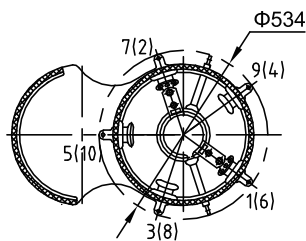
Linear Regulation Tap Selector



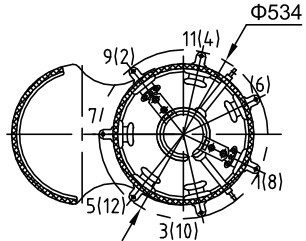
10070



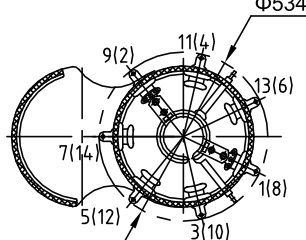
10090



10100

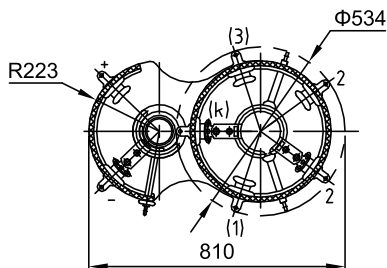


14120

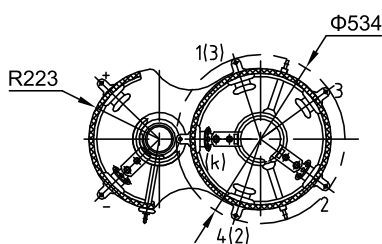


14140

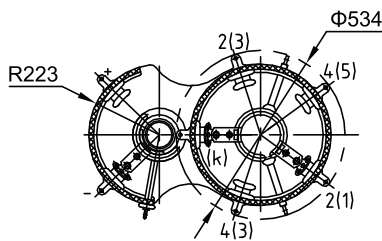
Reversing Tap Selector



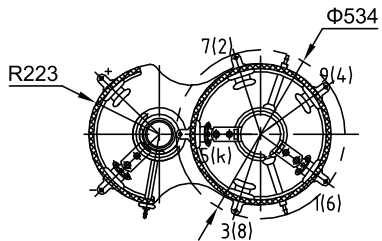
10051W



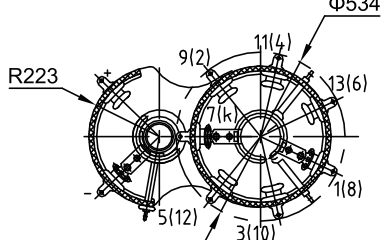
10071W



10091W

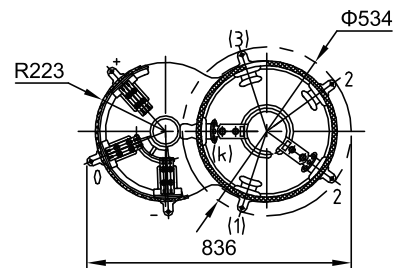


10191W

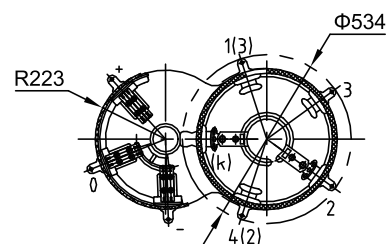


14271W

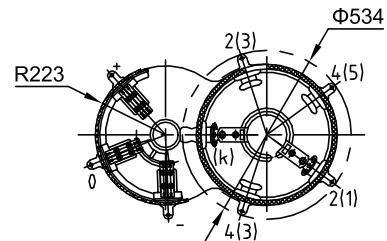
Coarse and Fine Tap Selector



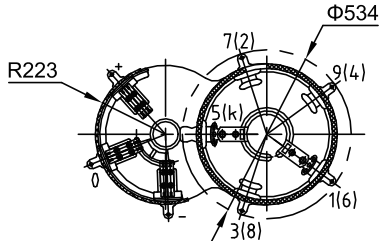
10051G



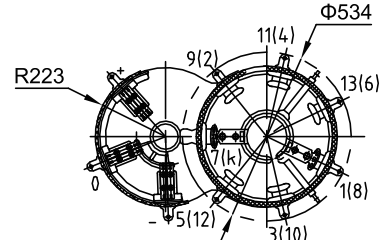
10071G



10091G



10191G

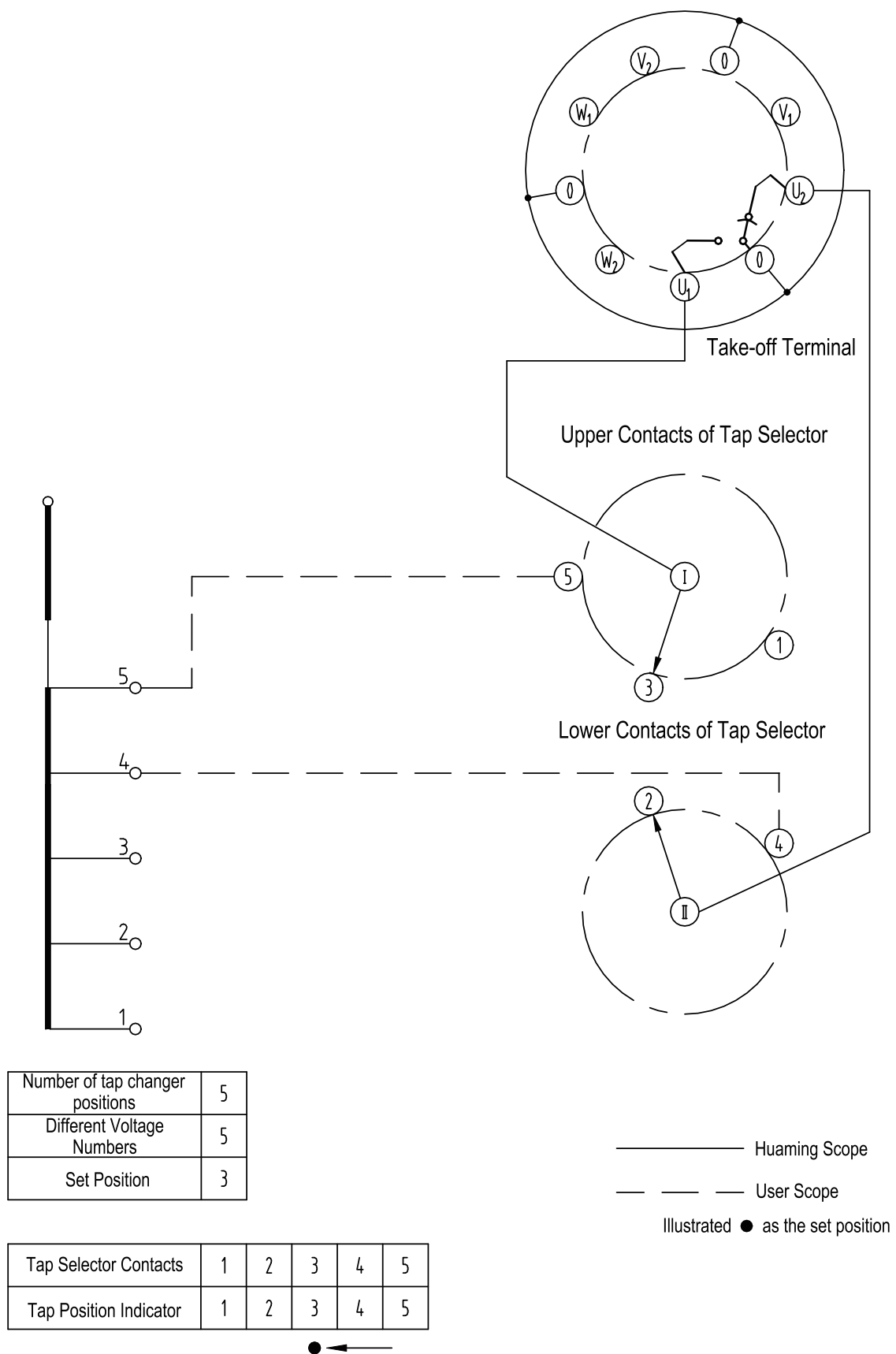


14271G

Note: The terminal designations should be referenced in the relevant wiring diagrams.

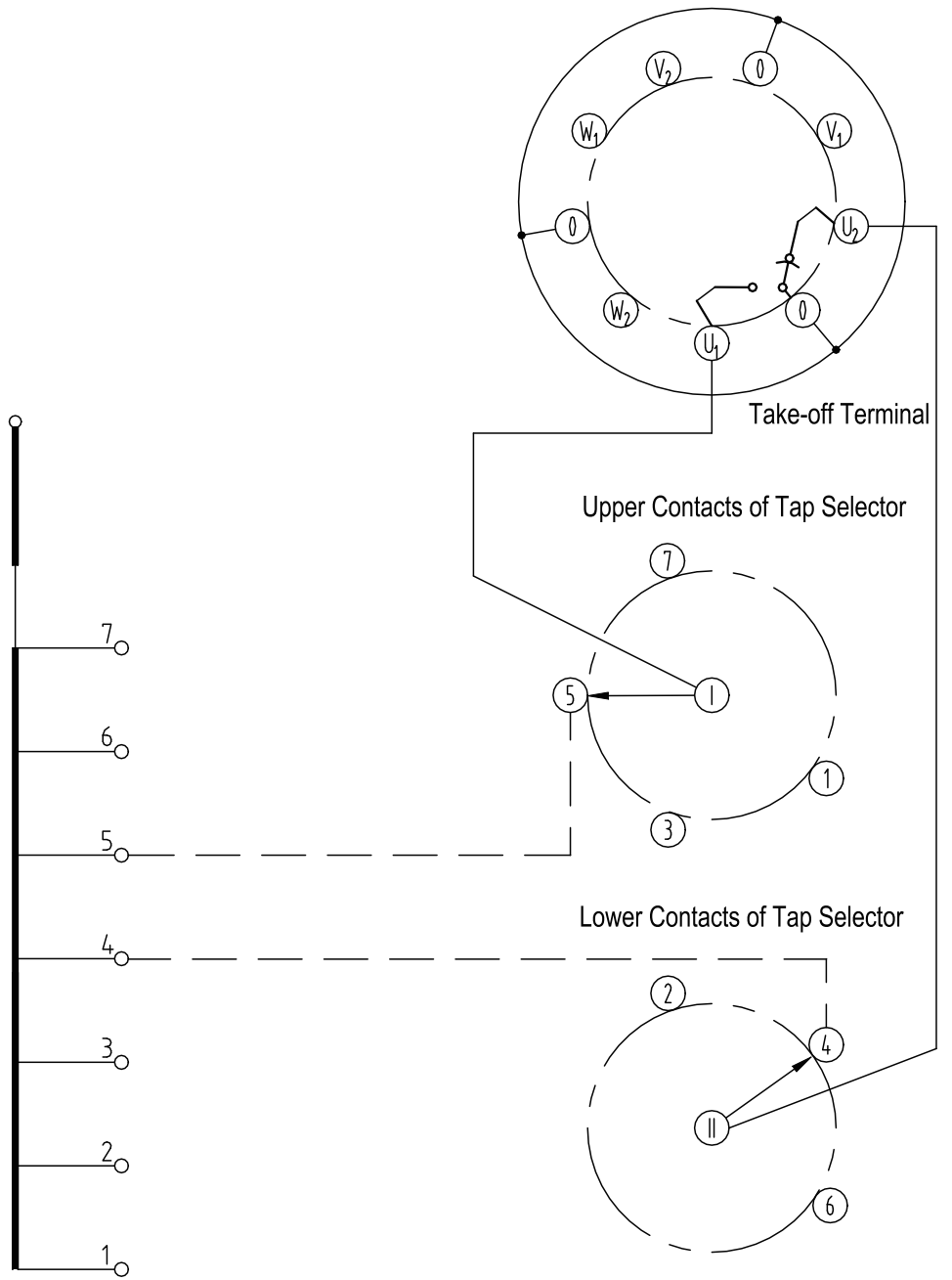
Tap Selector Terminal Designations:
X: Upper contact;
(X): Lower contact.

Appendix 14 SHZV (10050) operating position table and connection diagram





Appendix 15 SHZV (10070) operating position table and connection diagram



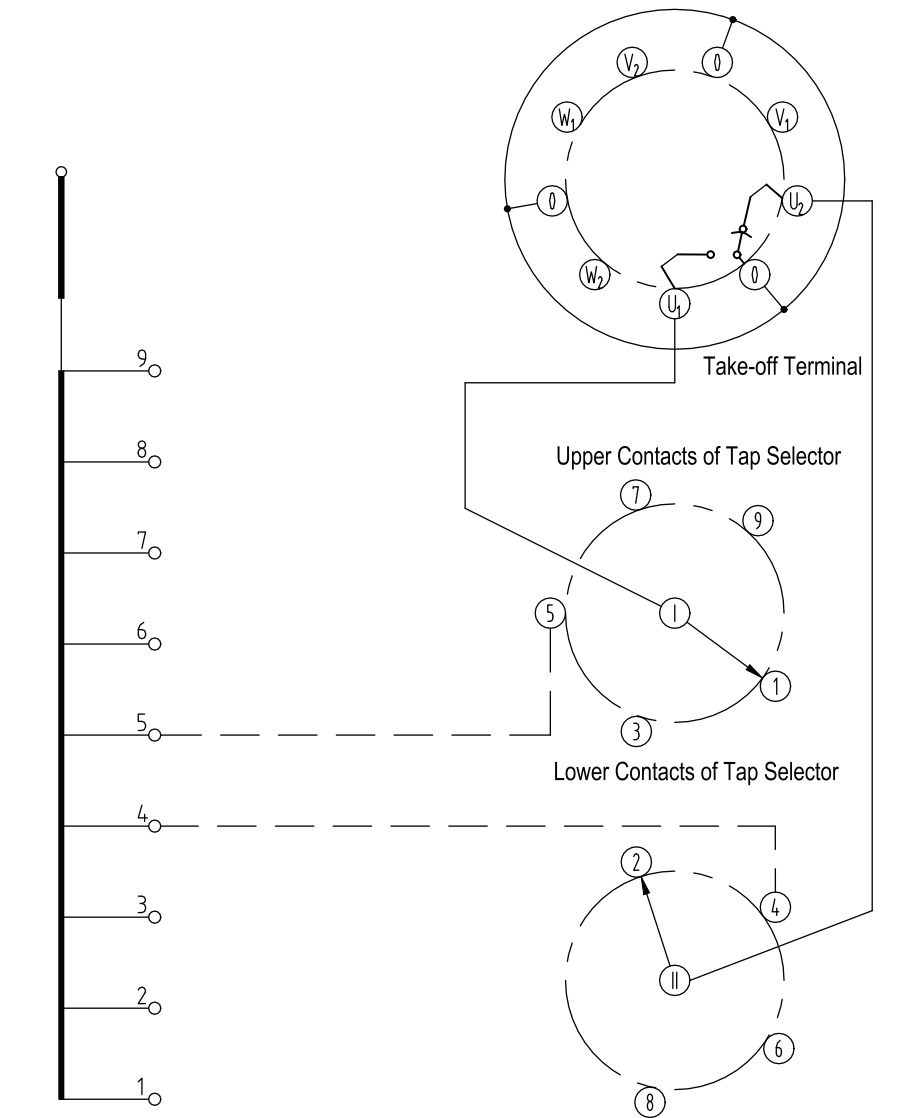
Number of tap changer positions	7
Different Voltage Numbers	7
Set Position	● 4

———— Huaming Scope
- - - - User Scope
Illustrated ● as the Set position

Tap Selector Contacts	1	2	3	4	5	6	7
Tap Position Indicator	1	2	3	4	5	6	7



Appendix 16 SHZV (10090) operating position table and connection diagram



Number of tap changer positions	9
Different Voltage Numbers	9
Set Position	● 5

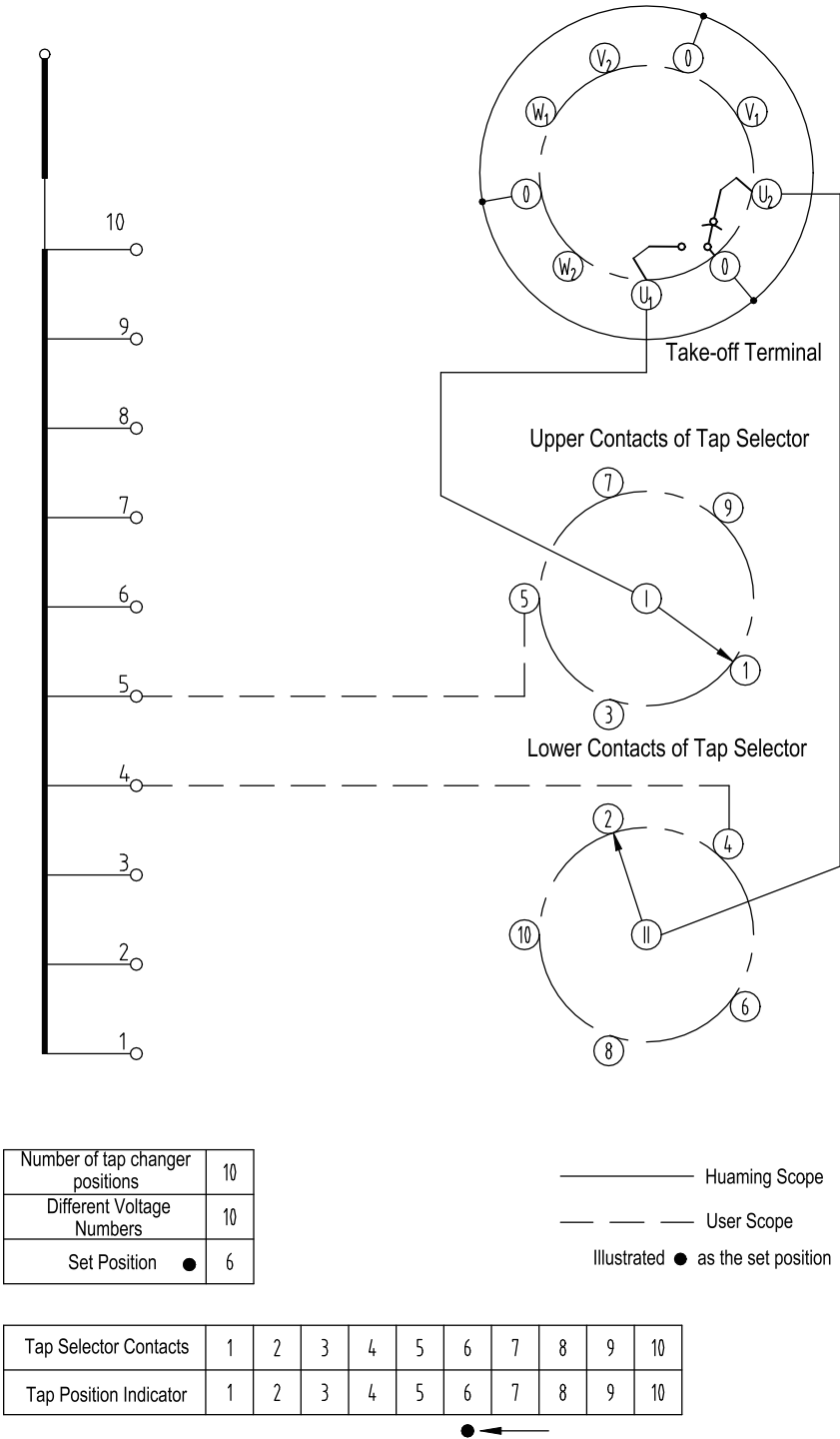
————— Huaming Scope
 - - - - - User Scope
 Illustrated ● as the set position

Tap Selector Contacts	1	2	3	4	5	6	7	8	9
Tap Position Indicator	1	2	3	4	5	6	7	8	9

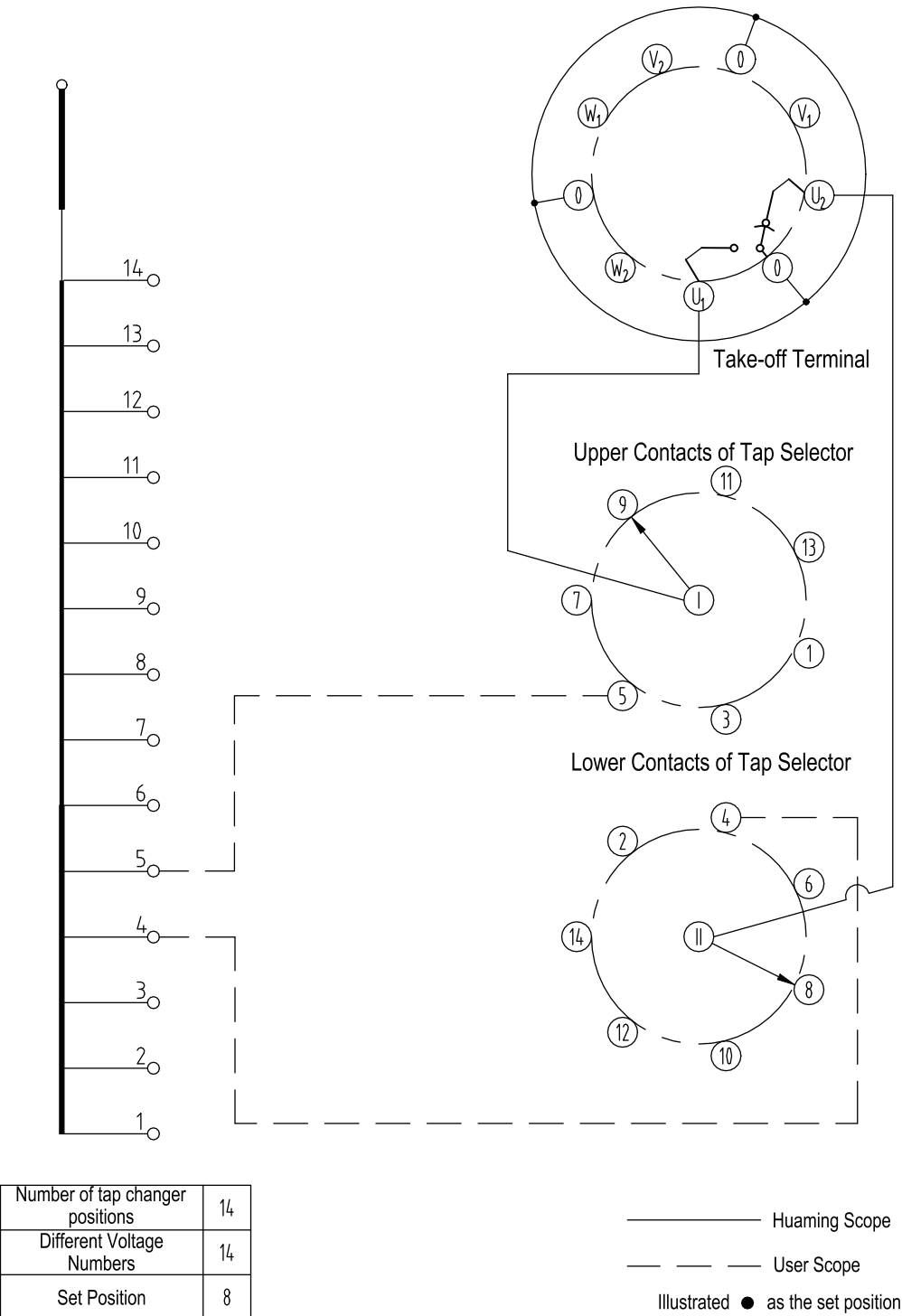
● ←



Appendix 17 SHZV (10100) operating position table and connection diagram



Appendix 18 SHZV (14140) operating position table and connection diagram

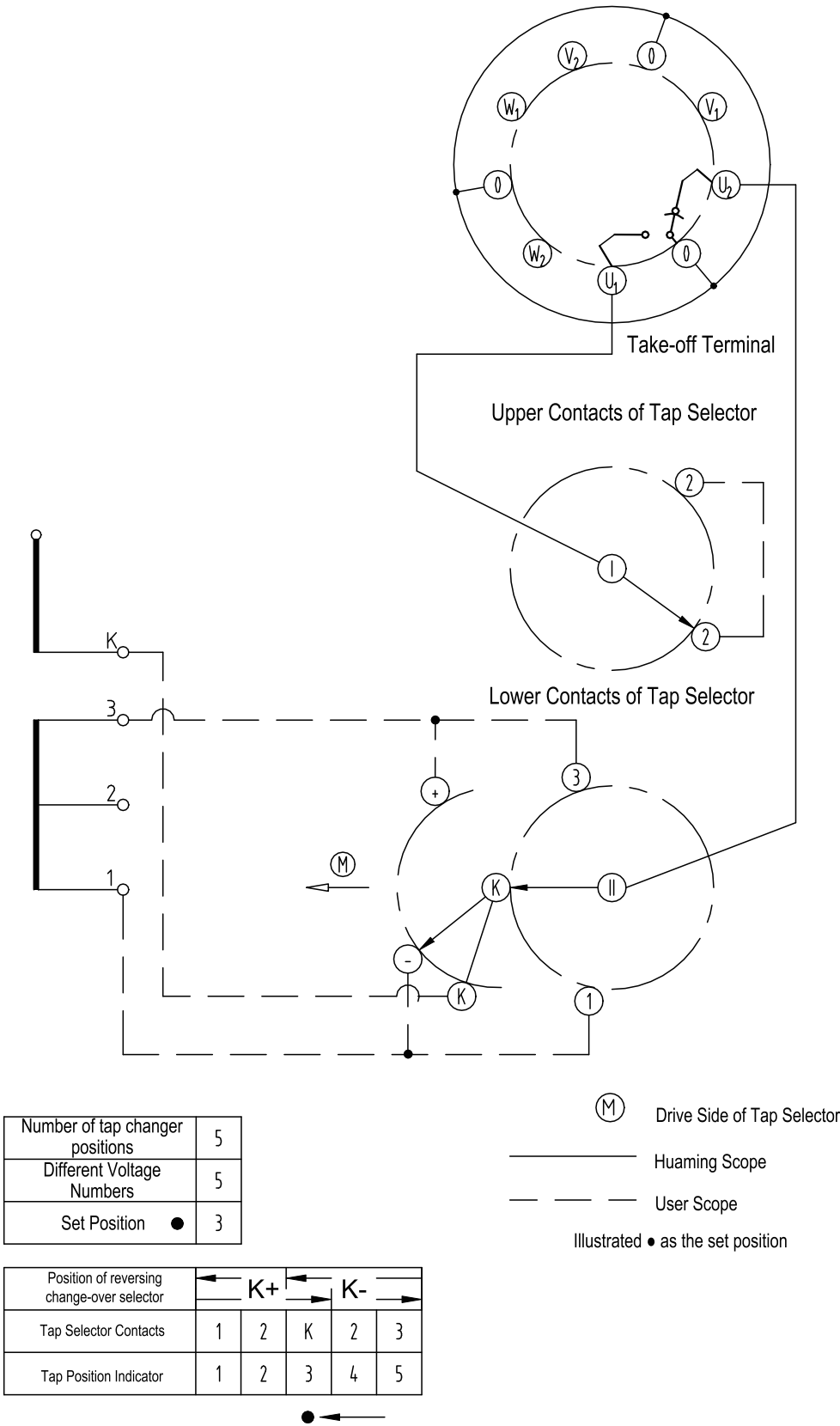


Tap Selector Contacts	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Tap Position Indicator	1	2	3	4	5	6	7	8	9	10	11	12	13	14

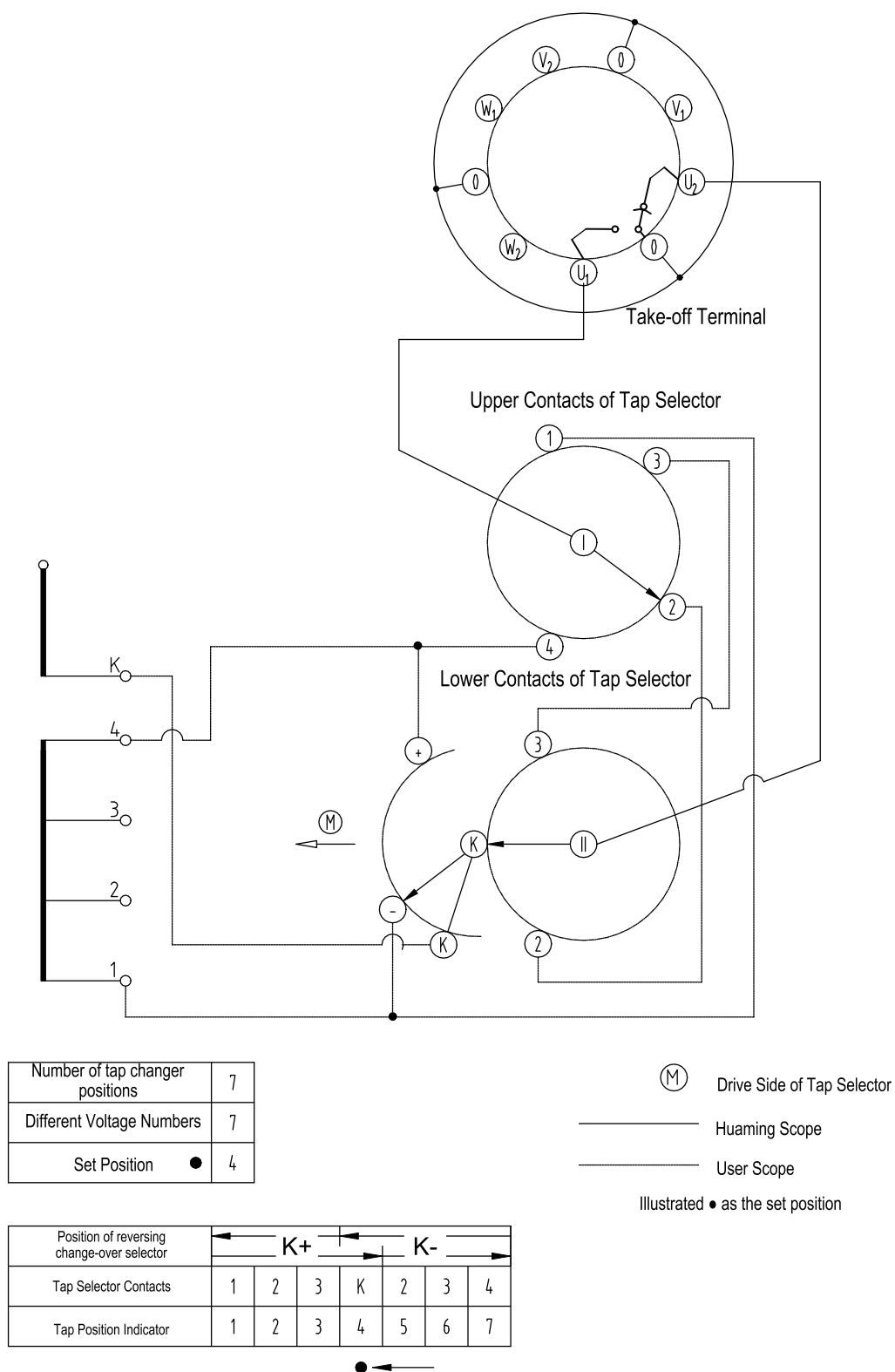




Appendix 19 SHZV (10051W) operating position table and connection diagram

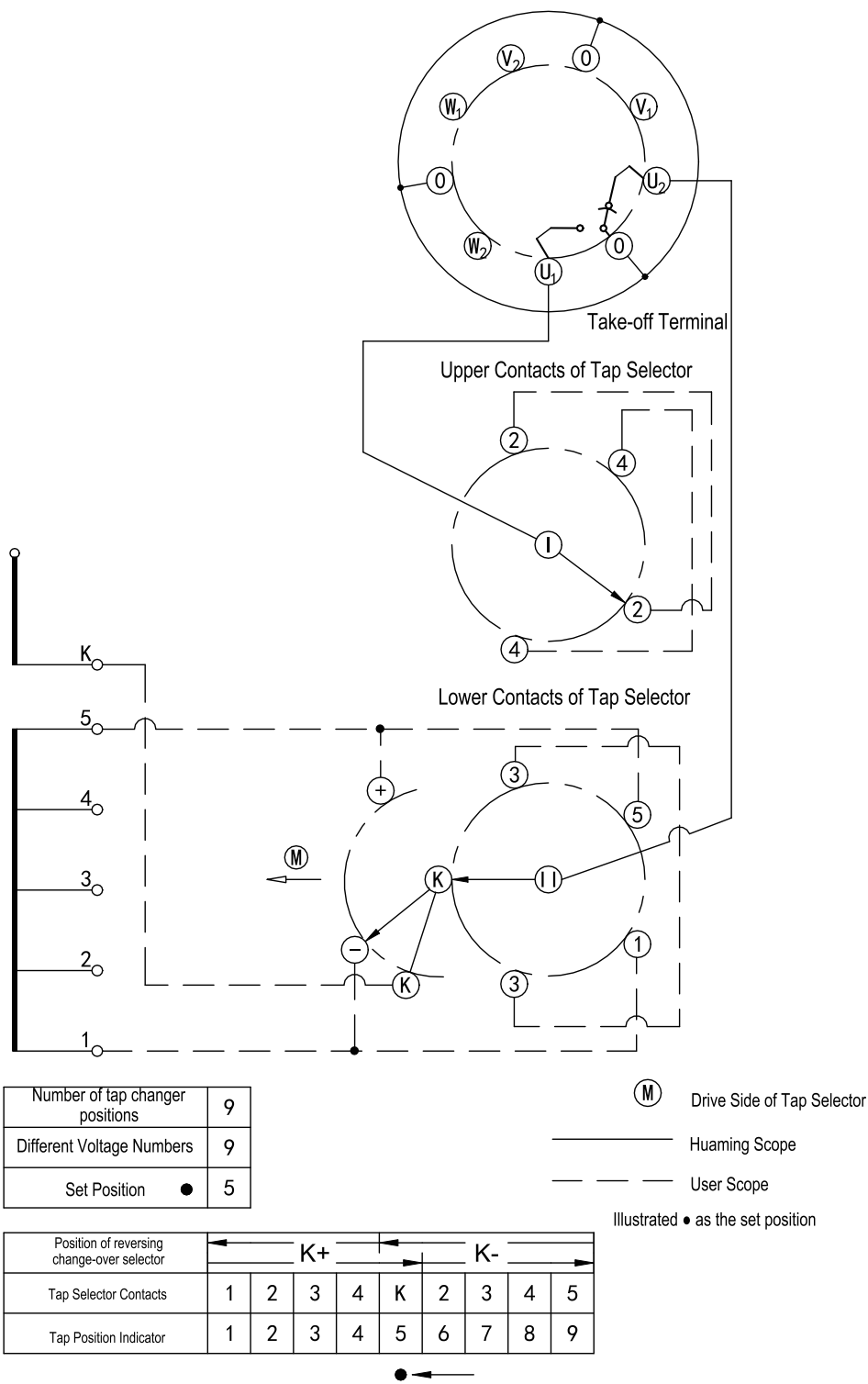


Appendix 20 SHZV (10071W) operating position table and connection diagram

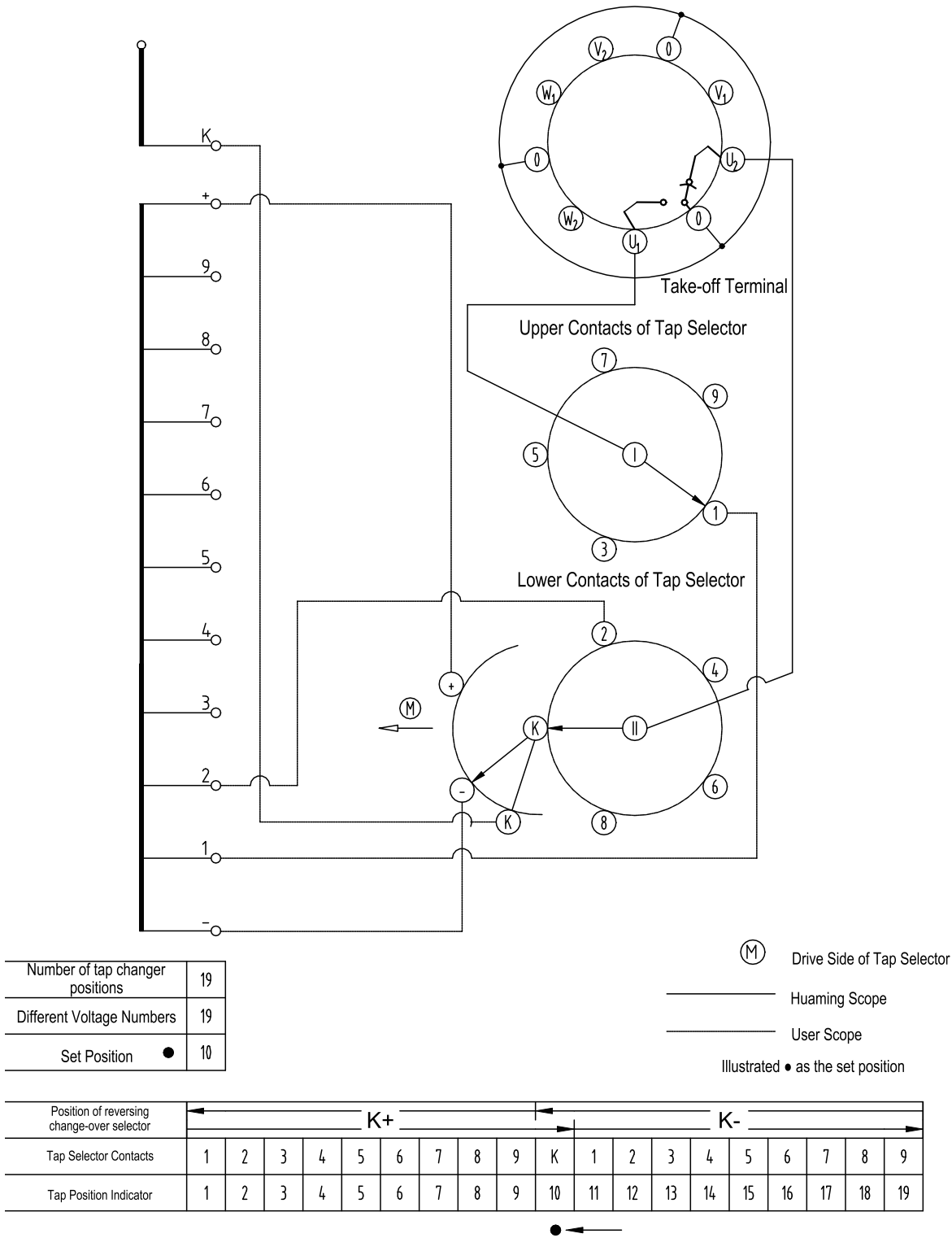




Appendix 21 SHZV (10091W) operating position table and connection diagram

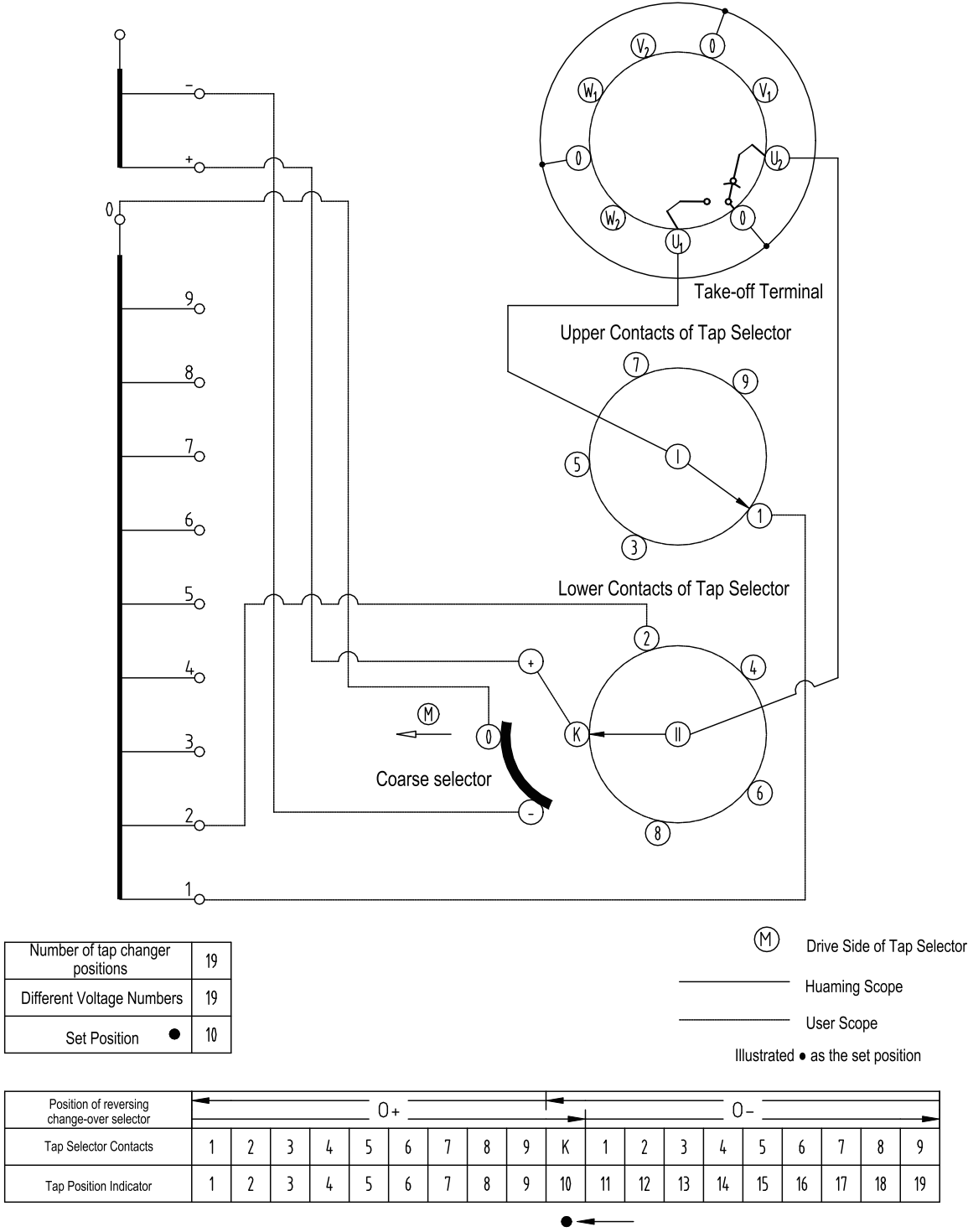


Appendix 22 SHZV (10191W) operating position table and connection diagram

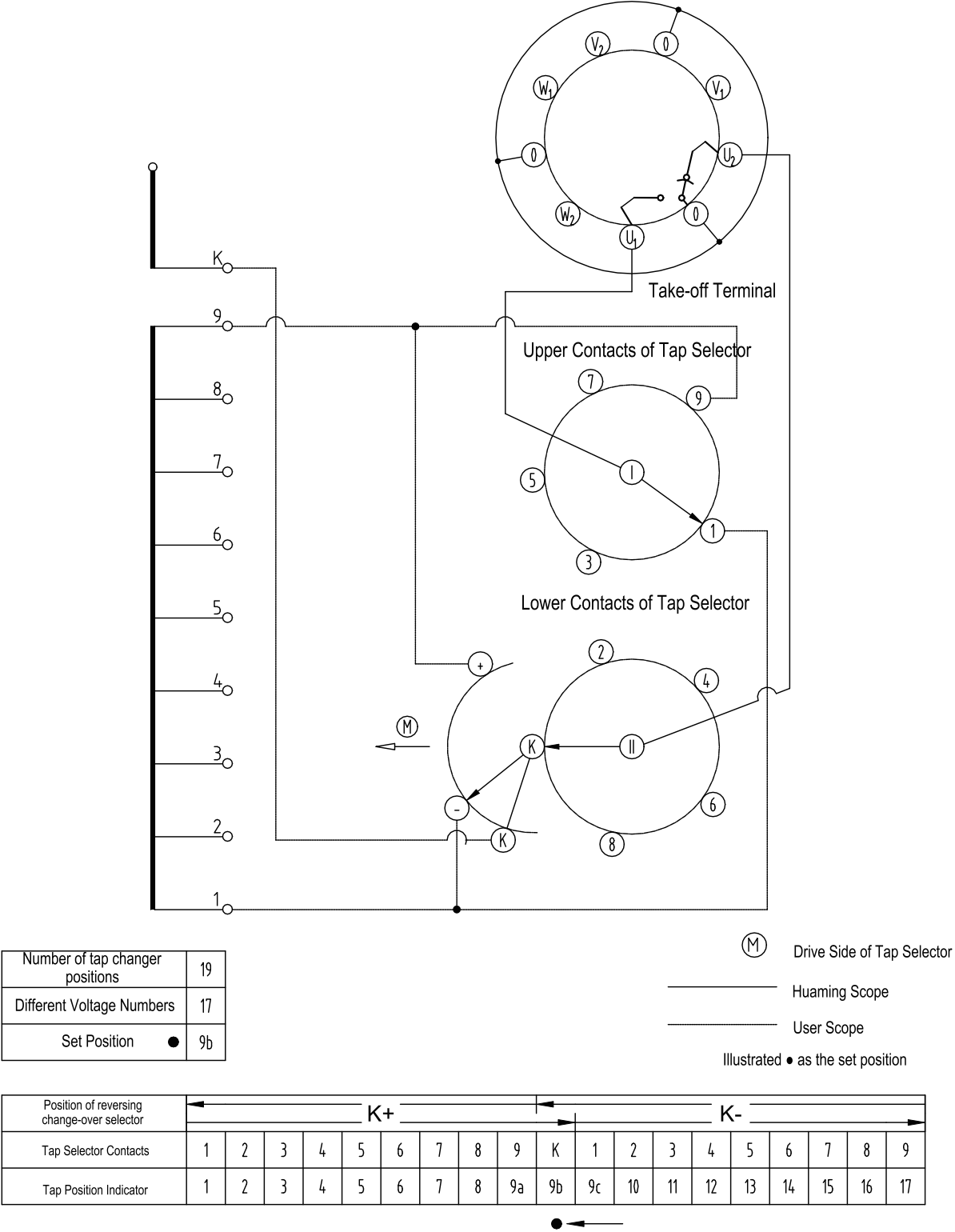




Appendix 23 SHZV (10191G) operating position table and connection diagram

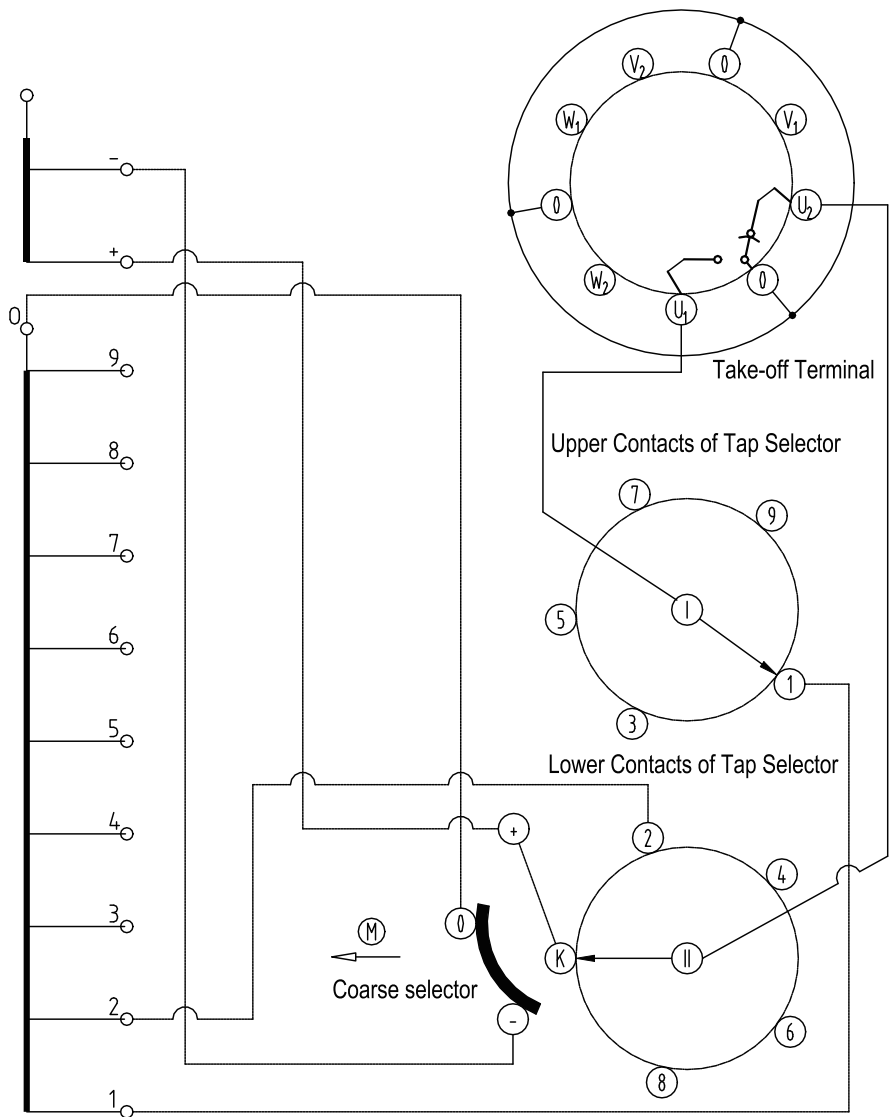


Appendix 24 SHZV (10193W) operating position table and connection diagram





Appendix 25 SHZV (10193G) operating position table and connection diagram



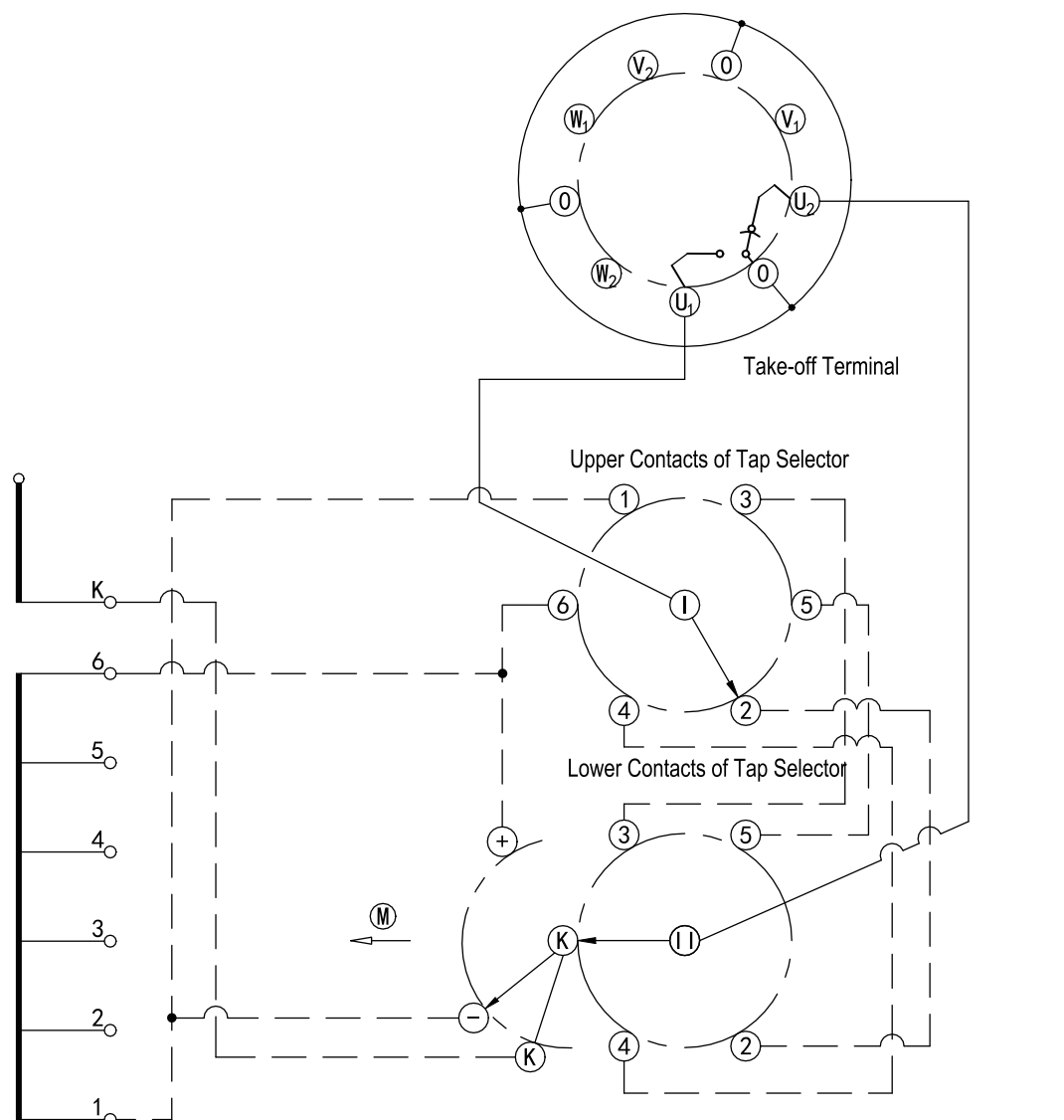
Number of tap changer positions	19
Different Voltage Numbers	17
Set Position	● 9b

- (M) Drive Side of Tap Selector
- Huaming Scope
- User Scope
- Illustrated ● as the set position

Position of reversing change-over selector	← 0+ →										← 0- →									
Tap Selector Contacts	1	2	3	4	5	6	7	8	9	K	1	2	3	4	5	6	7	8	9	
Tap Position Indicator	1	2	3	4	5	6	7	8	9a	9b	9c	10	11	12	13	14	15	16	17	



Appendix 26 SHZV (12111W) operating position table and connection diagram



Number of tap changer positions	11
Different Voltage Numbers	11
Set Position •	6

Ⓜ Drive Side of Tap Selector

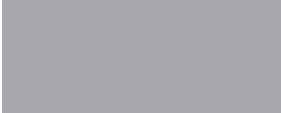
———— Huaming Scope

- - - - User Scope

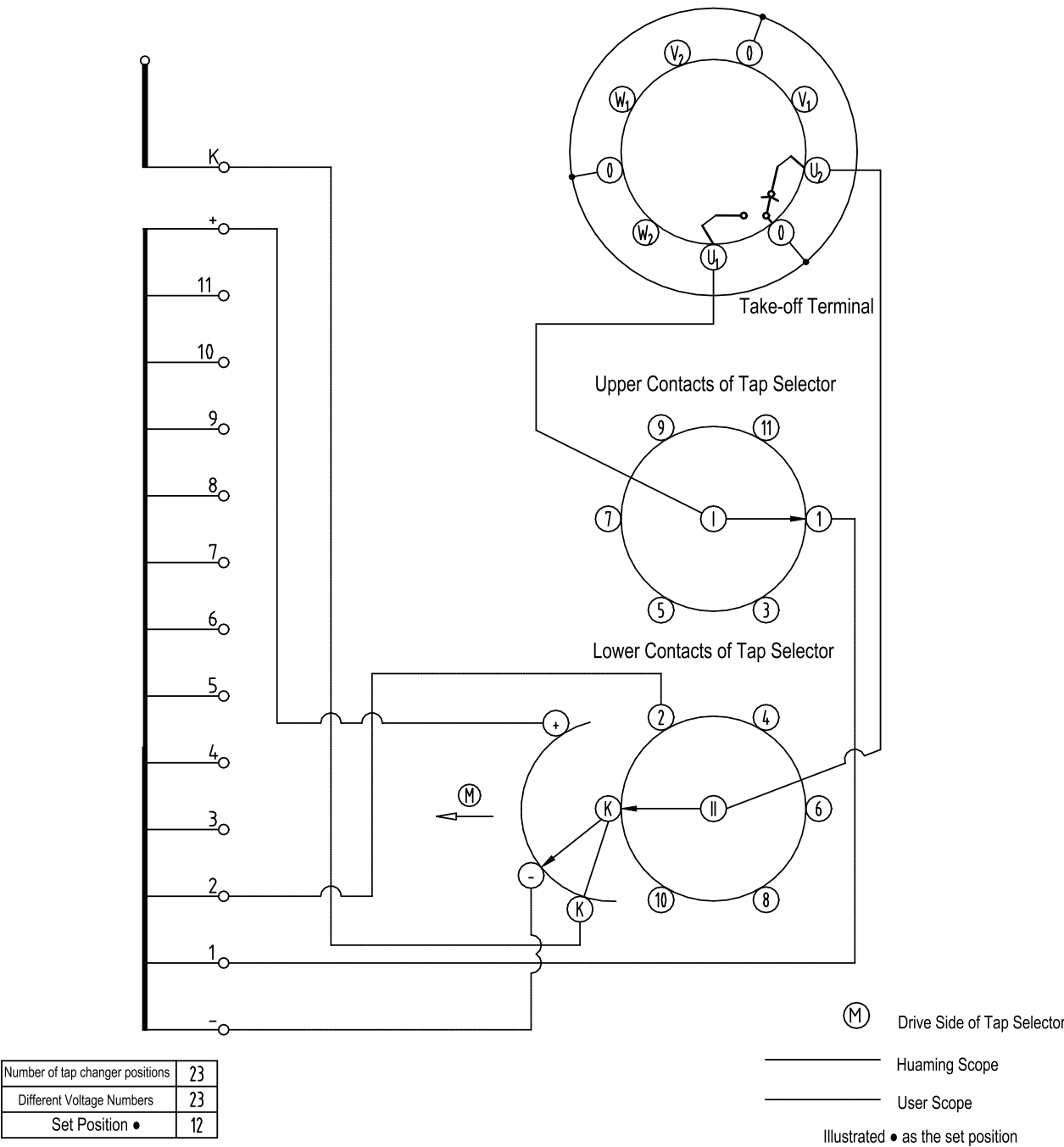
Illustrated • as the set position

Position of reversing change-over selector	K+					K-					
Tap Selector Contacts	1	2	3	4	5	K	2	3	4	5	6
Tap Position Indicator	1	2	3	4	5	6	7	8	9	10	11





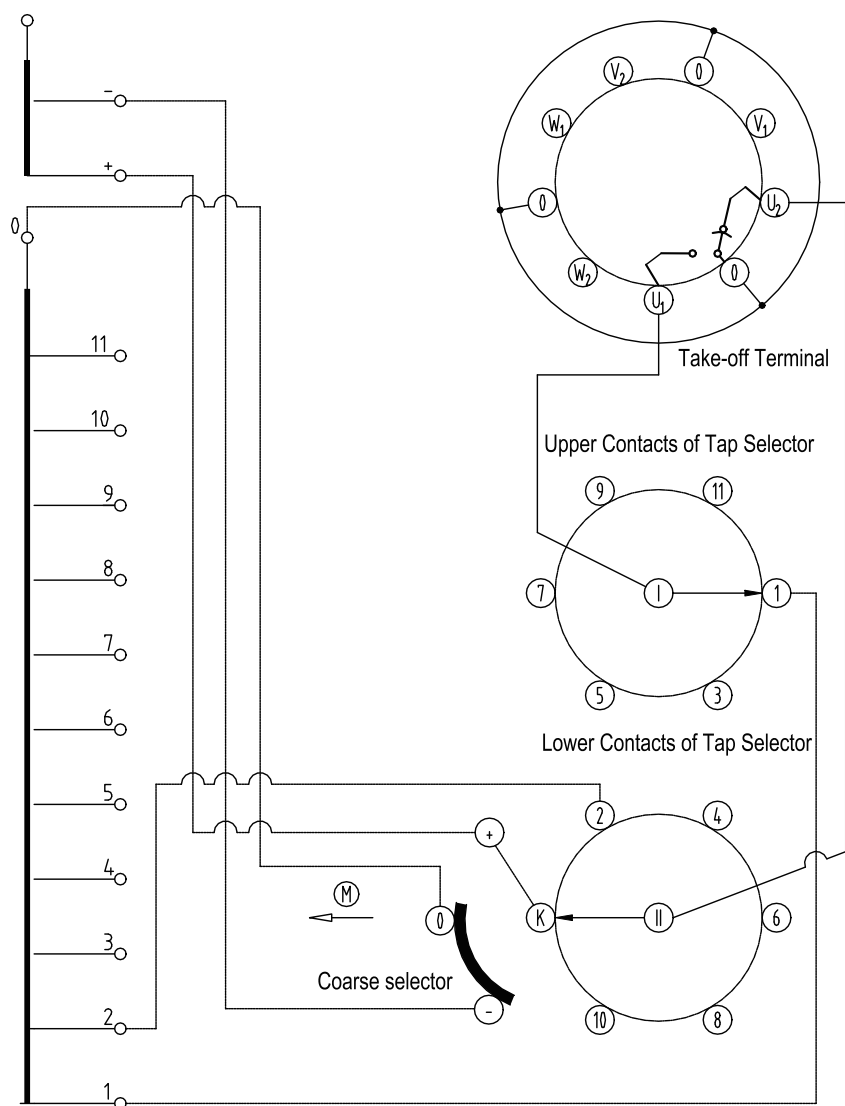
Appendix 27 SHZV (12231W) operating position table and connection diagram



Position of reversing change-over selector	← K+ →											← K- →											
Tap Selector Contacts	1	2	3	4	5	6	7	8	9	10	11	K	1	2	3	4	5	6	7	8	9	10	11
Tap Position Indicator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23

● ←

Appendix 28 SHZV (12231G) operating position table and connection diagram



Number of tap changer positions	23
Different Voltage Numbers	23
Set Position	● 12

(M) Drive Side of Tap Selector

— Huaming Scope

— User Scope

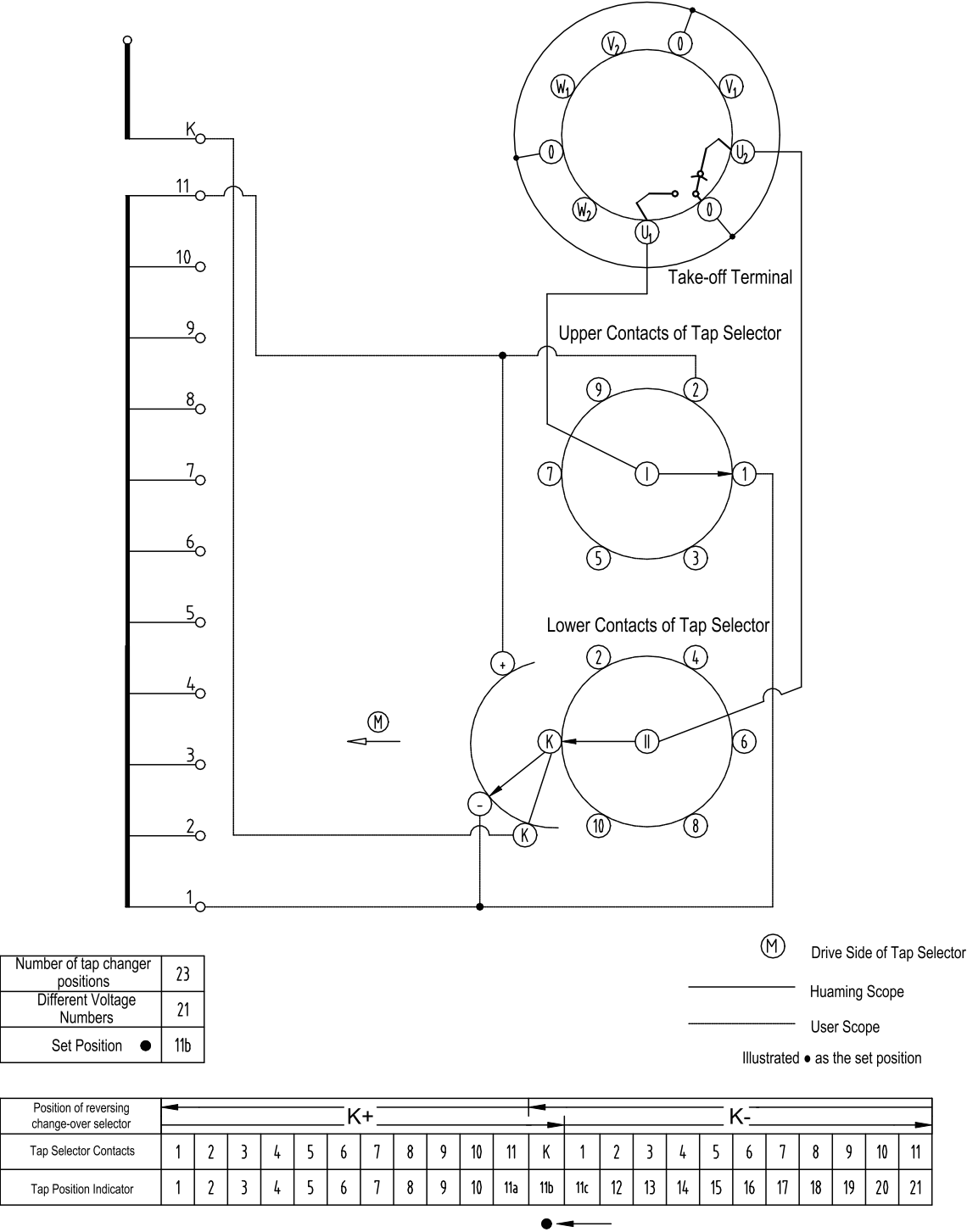
Illustrated ● as the set position

Position of reversing change-over selector																							
Tap Selector Contacts	1	2	3	4	5	6	7	8	9	10	11	K	1	2	3	4	5	6	7	8	9	10	11
Tap Position Indicator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23

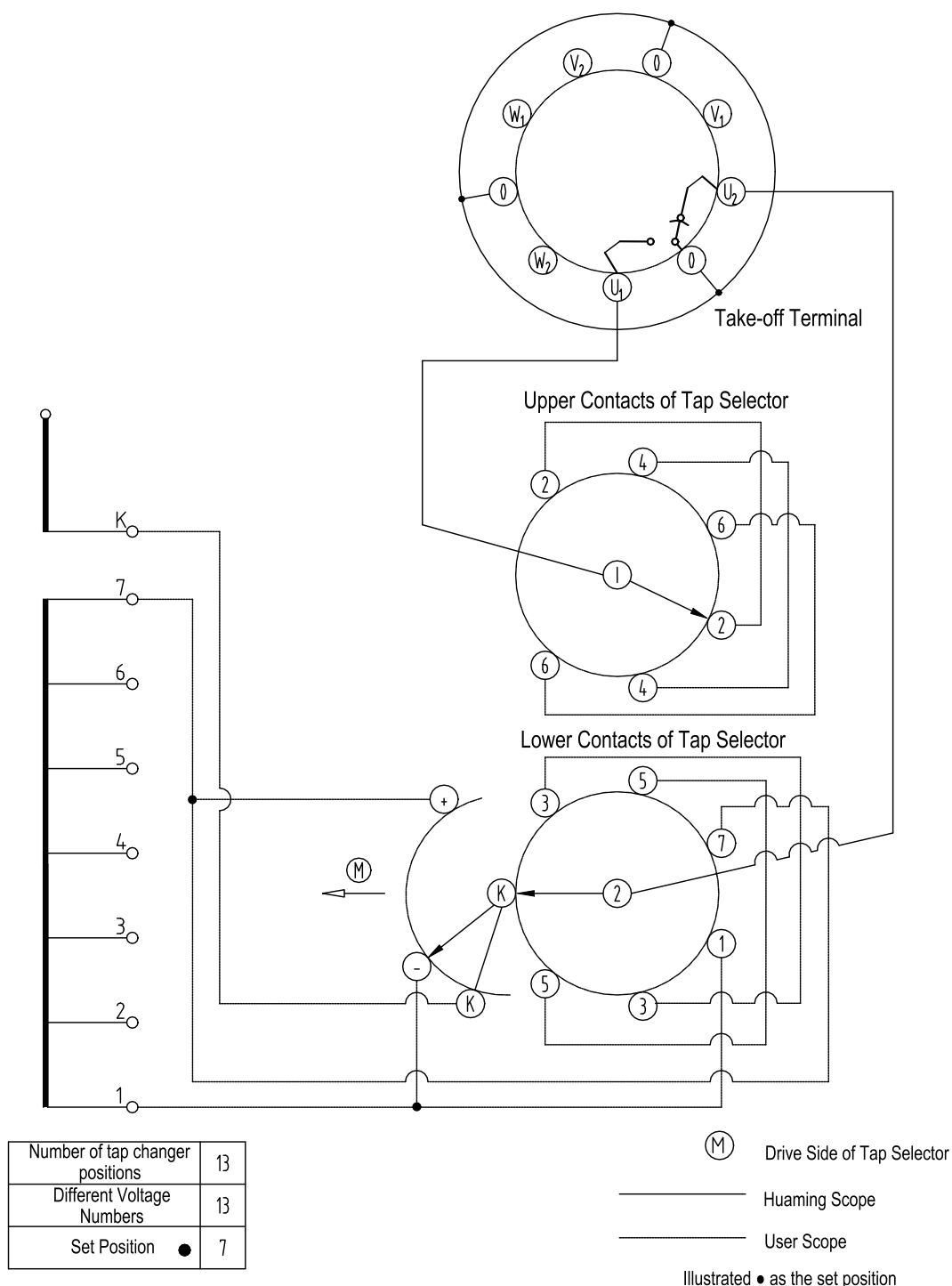
● ←



Appendix 29 SHZV (12233W) operating position table and connection diagram

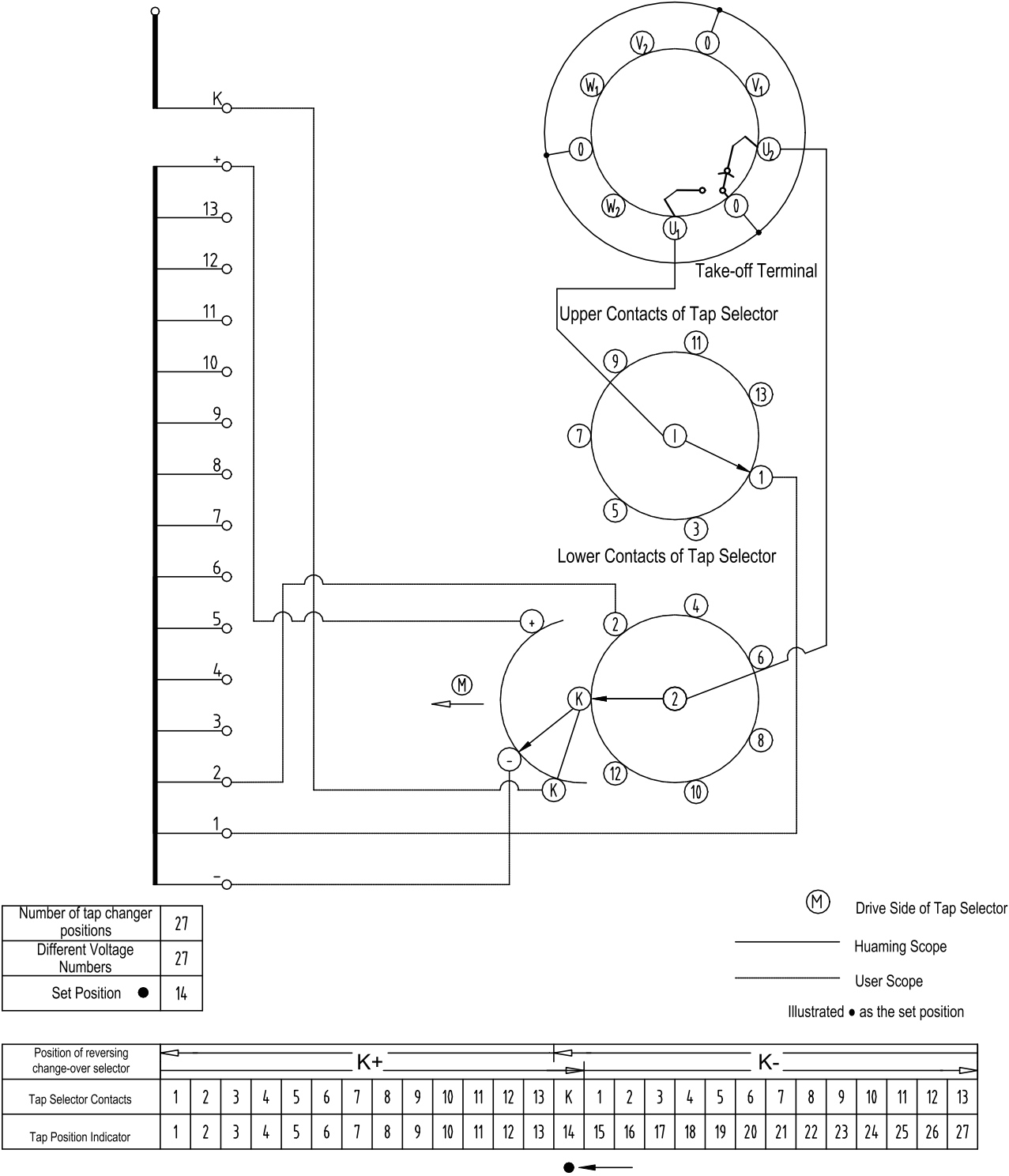


Appendix 30 SHZV (14131W) operating position table and connection diagram

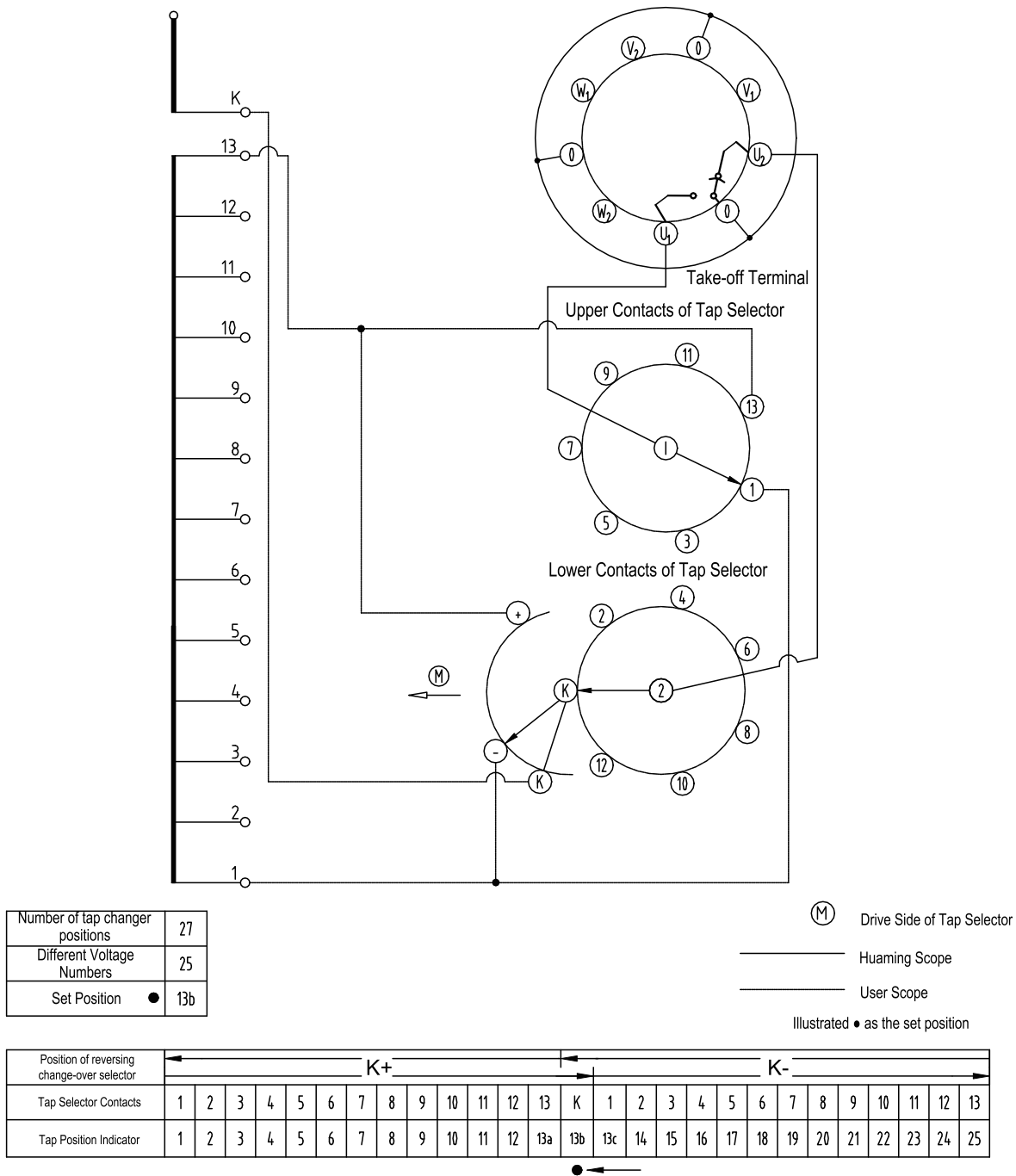




Appendix 31 SHZV (14271W) operating position table and connection diagram

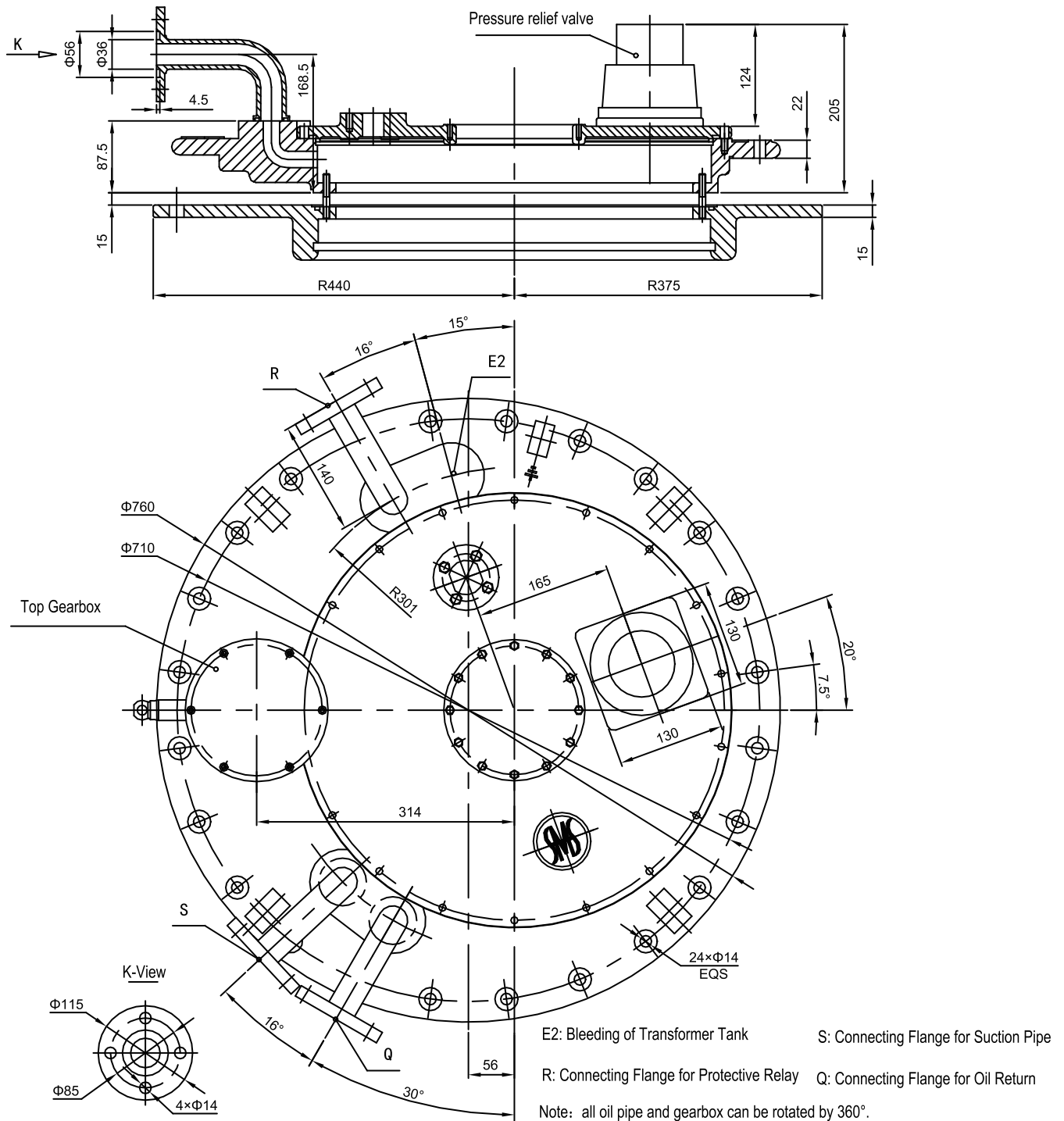


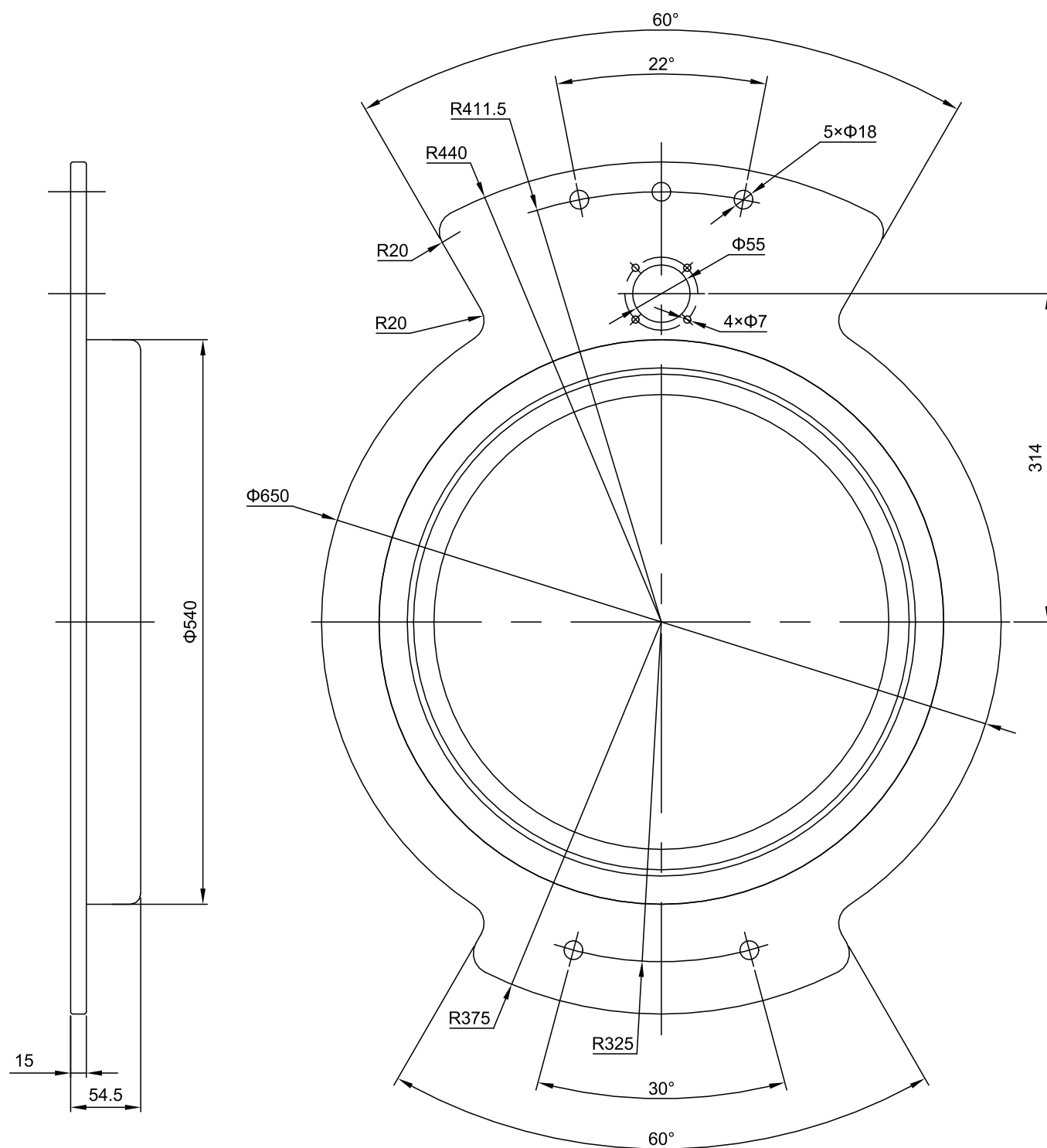
Appendix 32 SHZV (14273W) operating position table and connection diagram

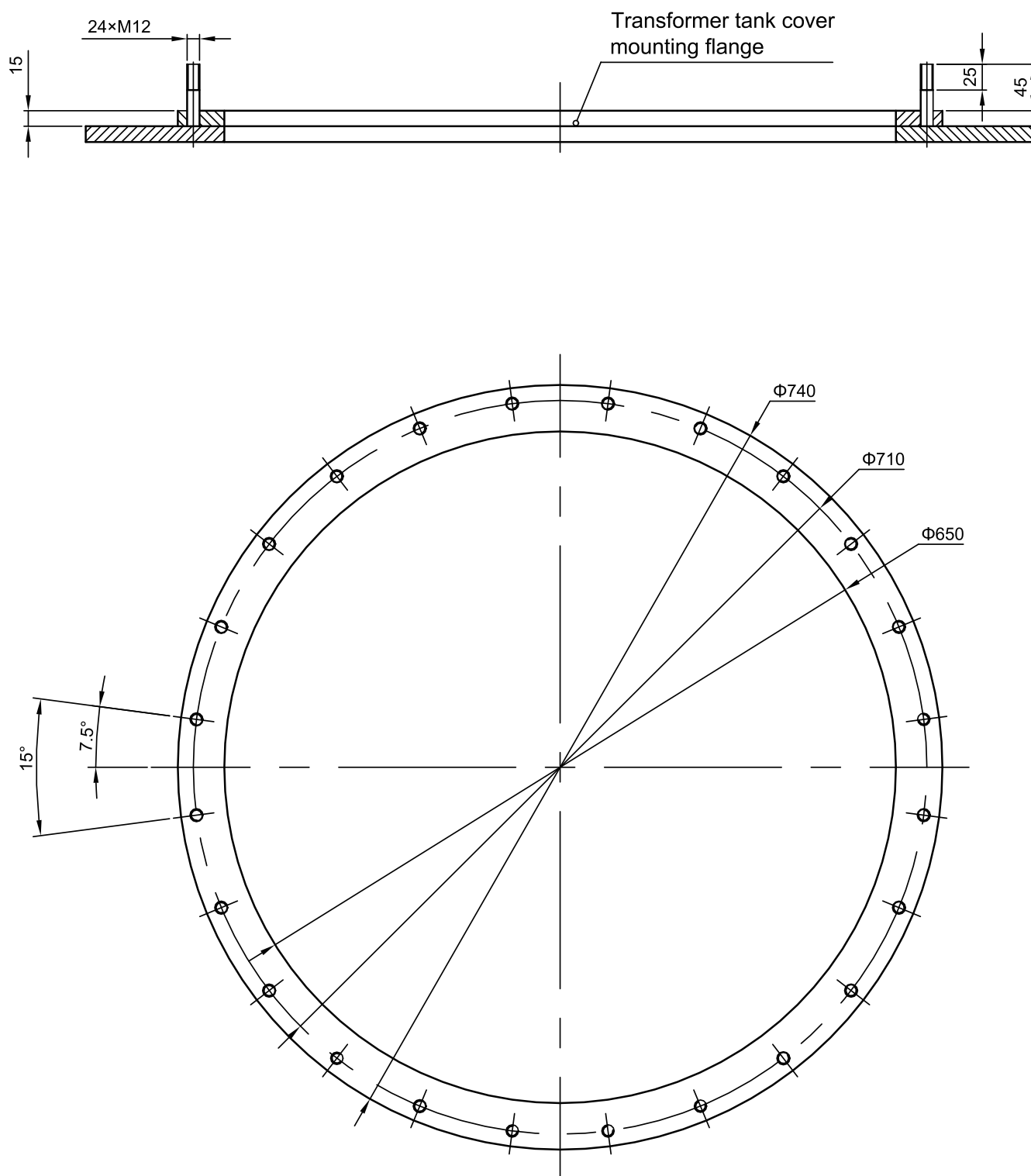




Appendix 34 SHZV bell-type mounting flange with pressure relief valve overall dimension, cylinder tap selector





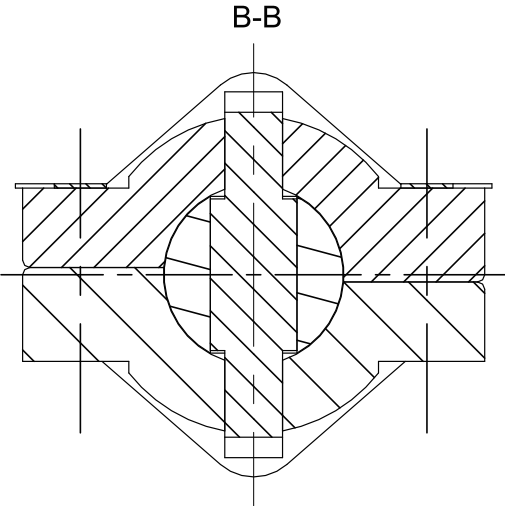
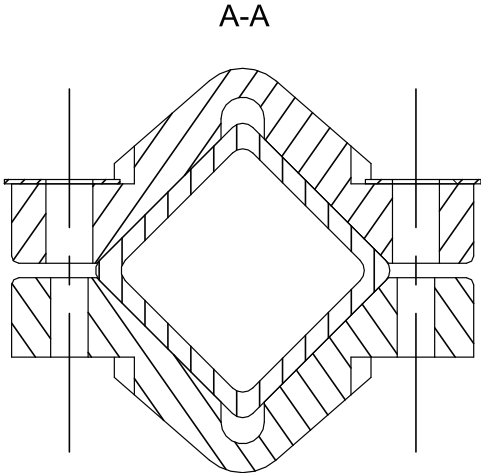
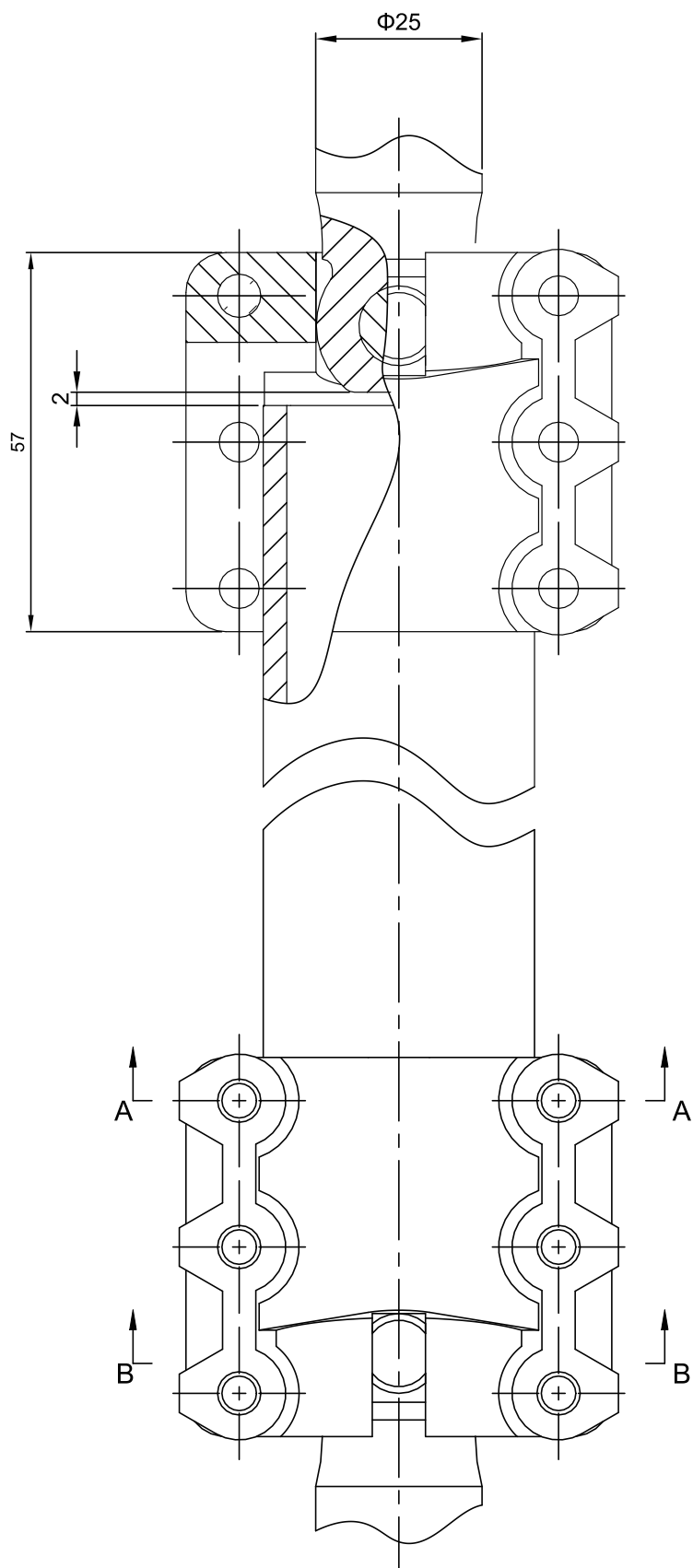
Appendix 36 Transformer mounting flange for SHZV overall dimension

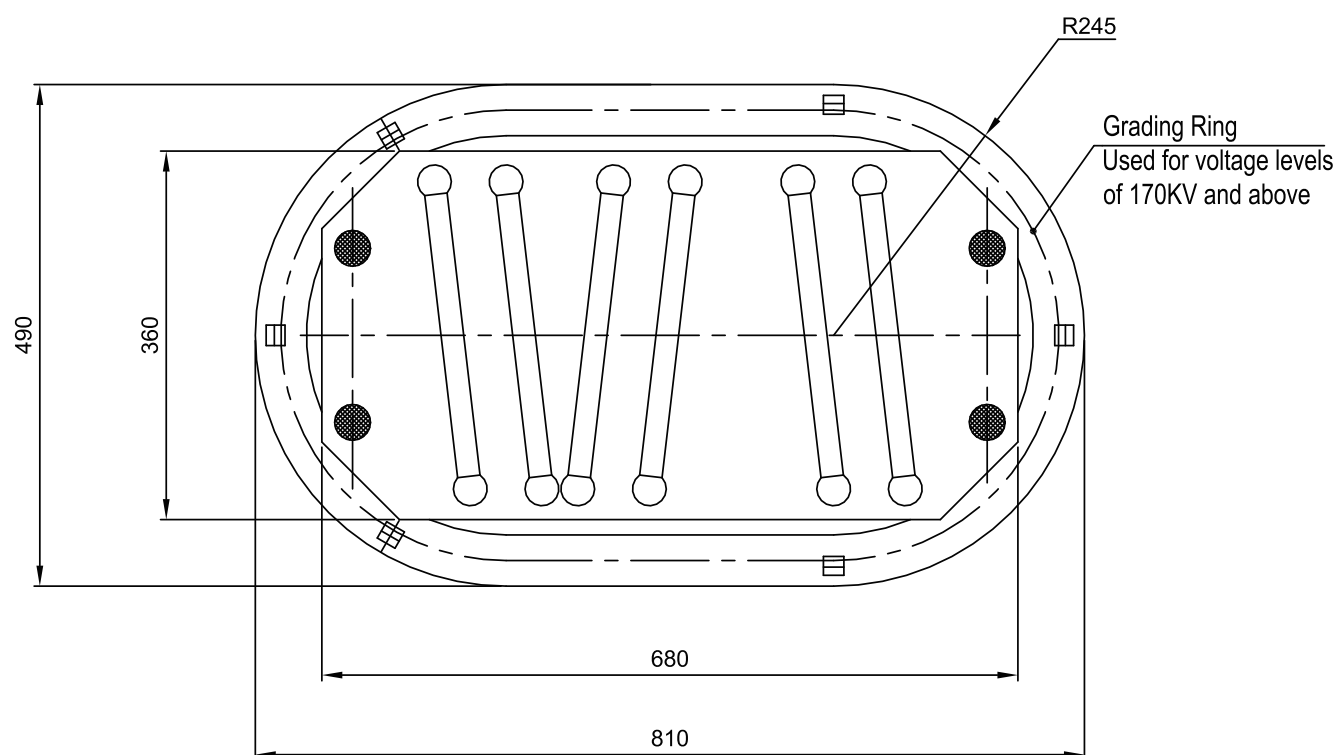
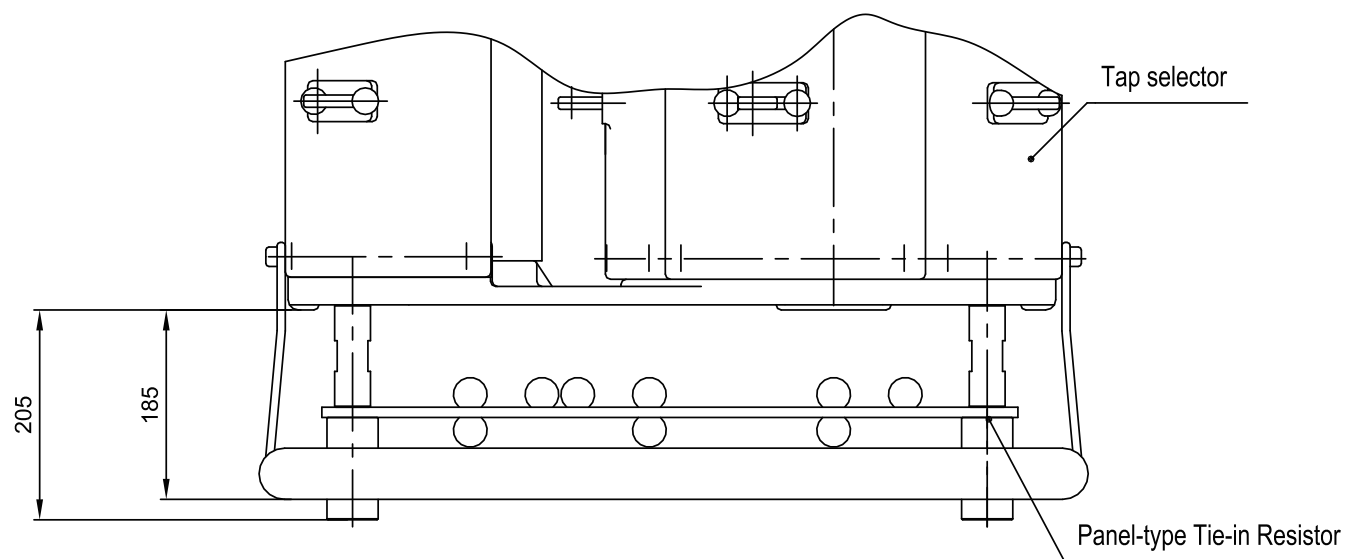




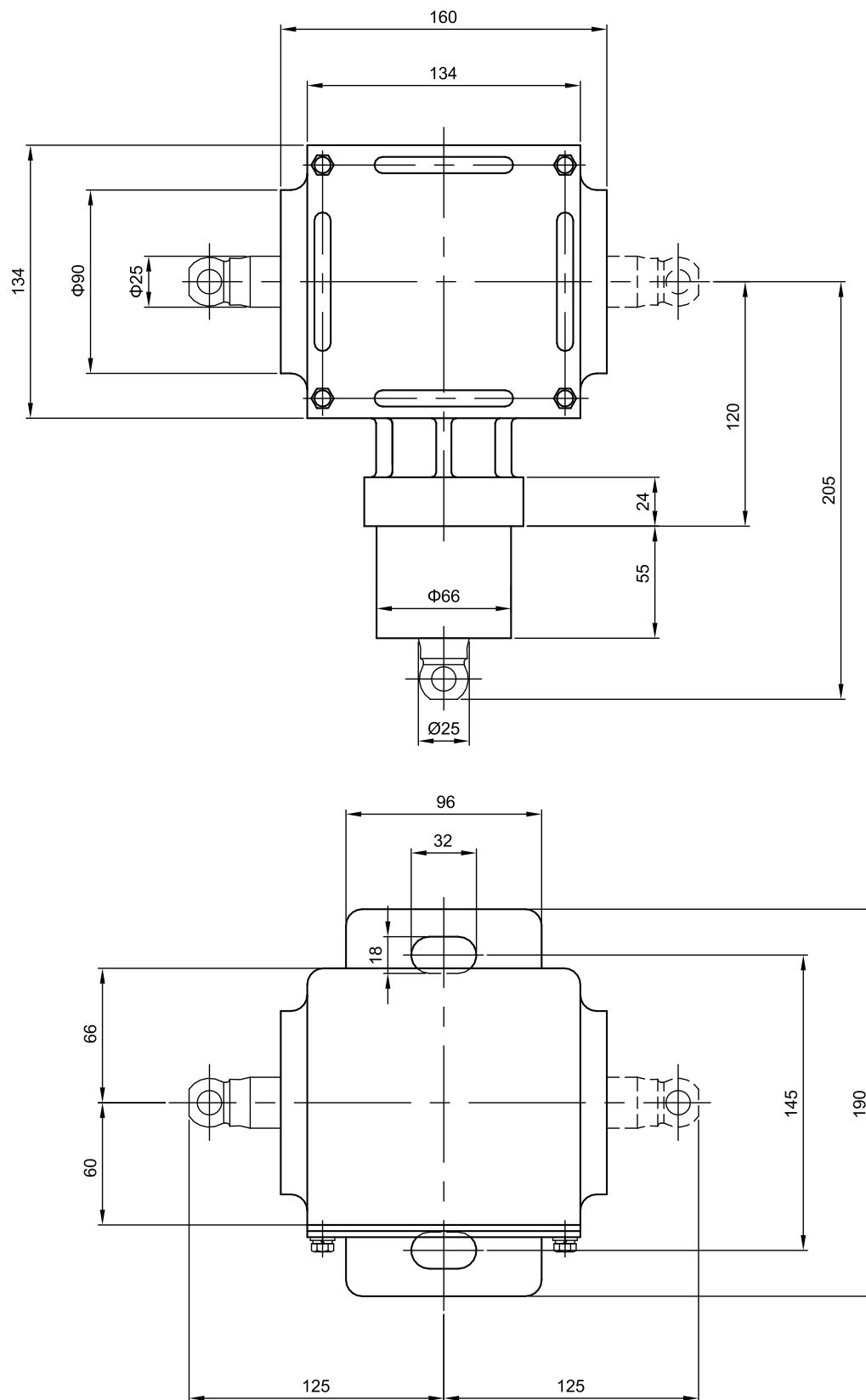


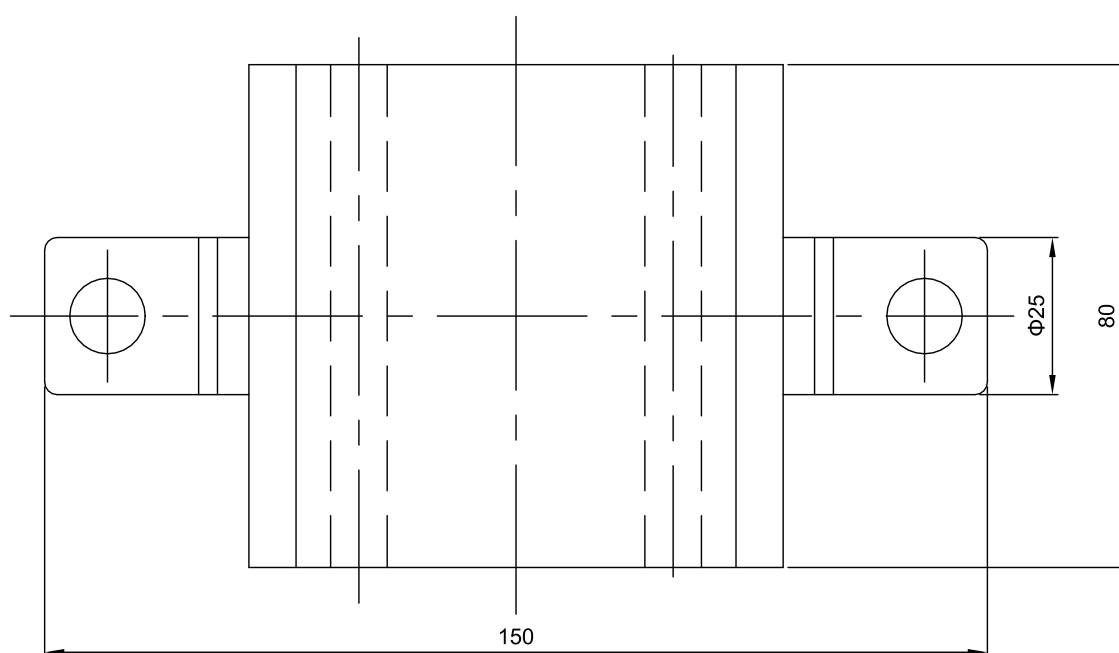
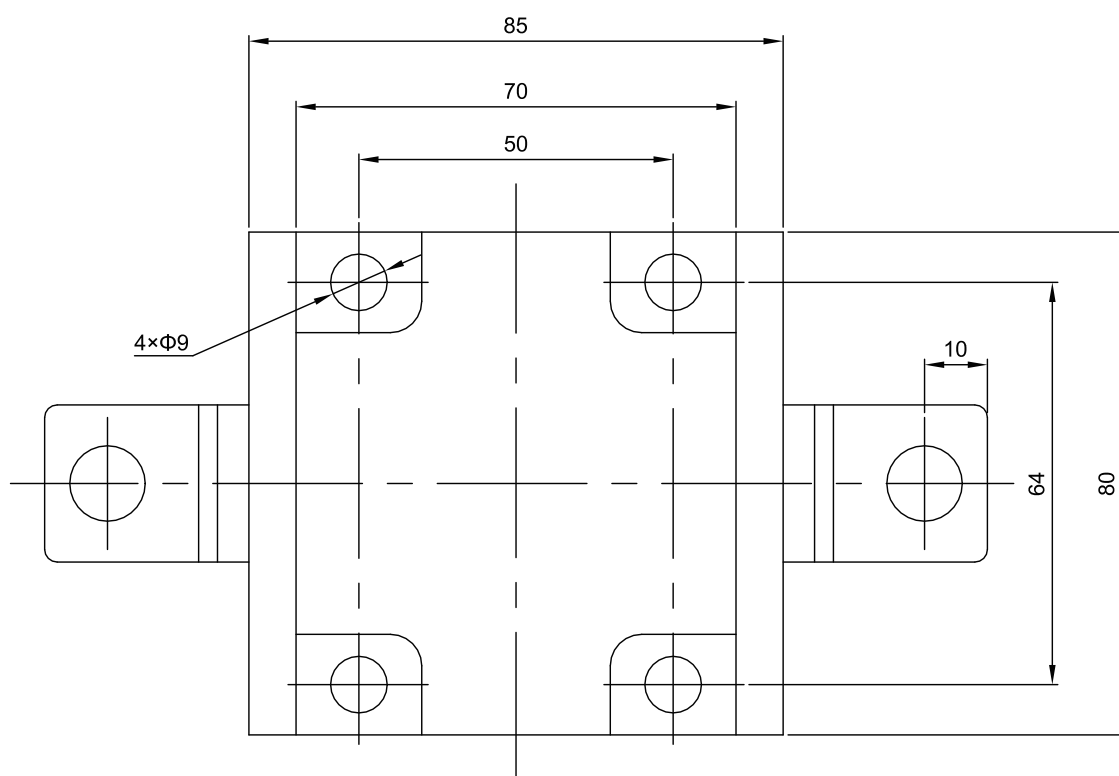
Appendix 39 Schematic drawing for connecting of horizontal shaft and vertical shaft



Appendix 40 Tie-in resistor mounting for overall dimension

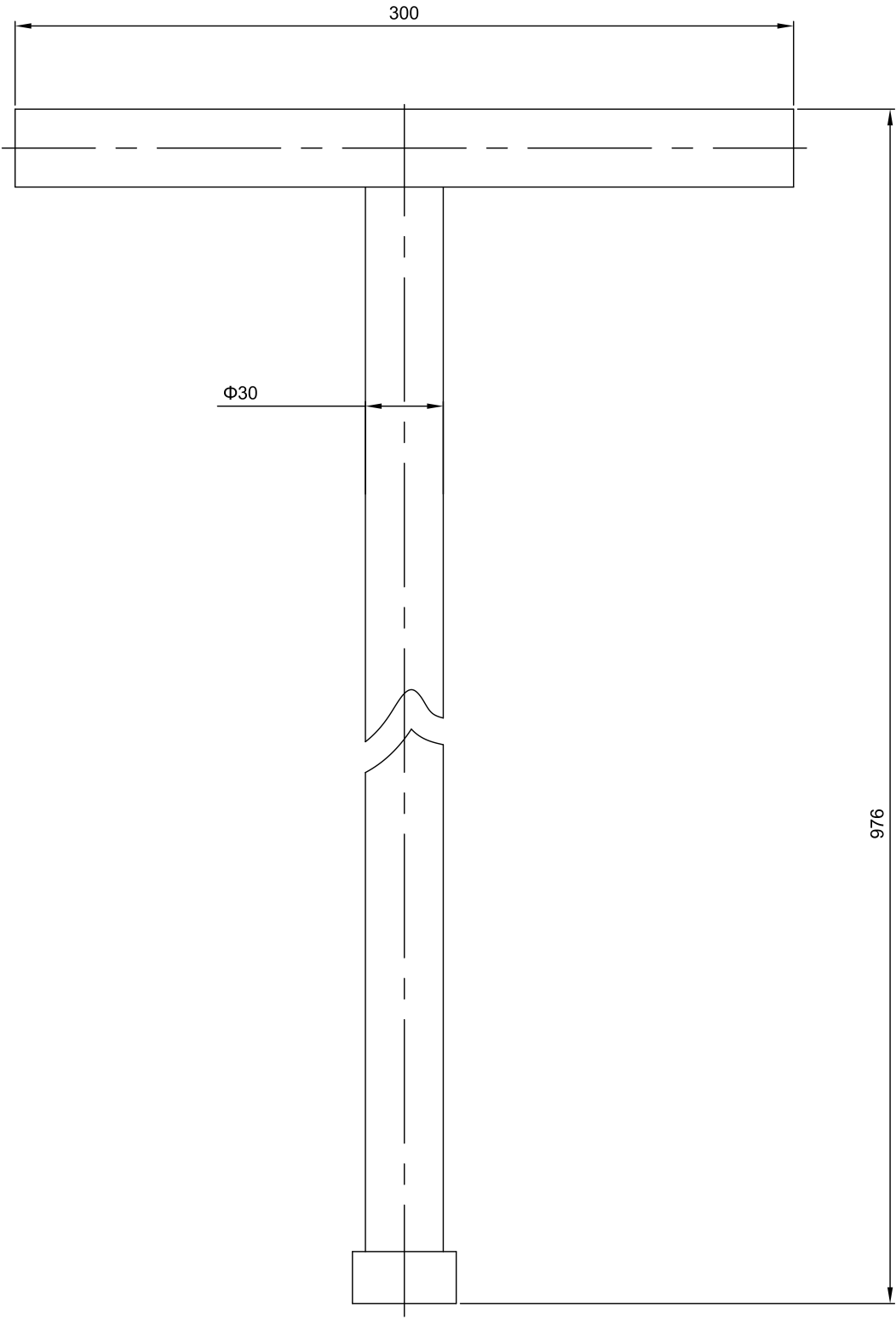
Appendix 41 Bevel gearbox, overall dimension



Appendix 42 Middle supporting box, overall dimension

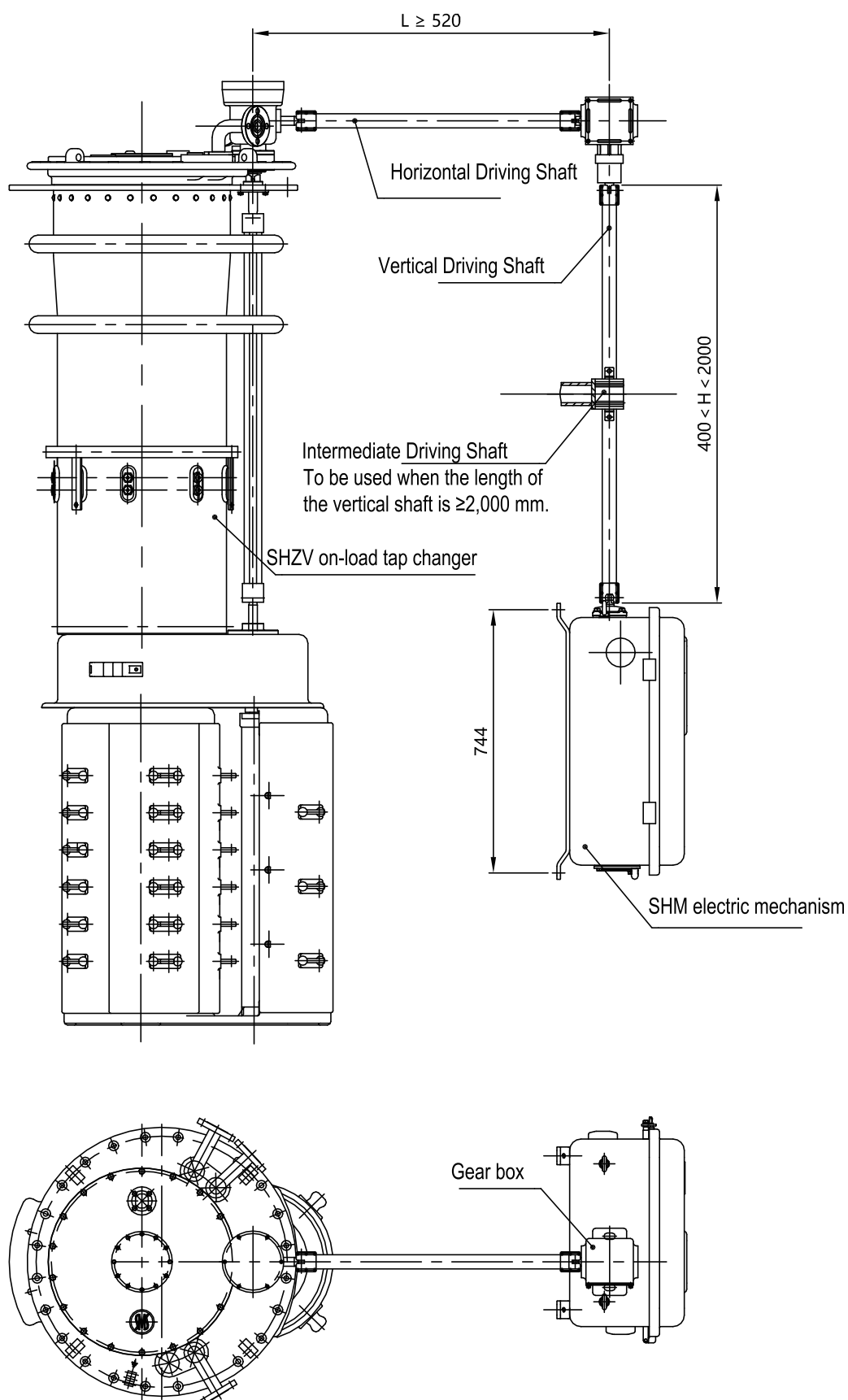


Appendix 43 Transformer oil pressure relief valve operating lever

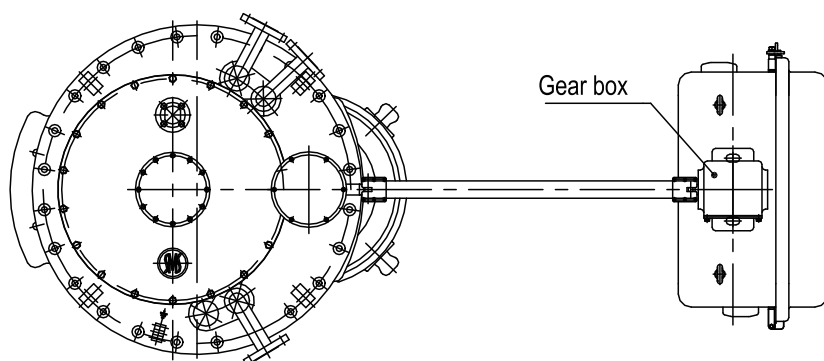
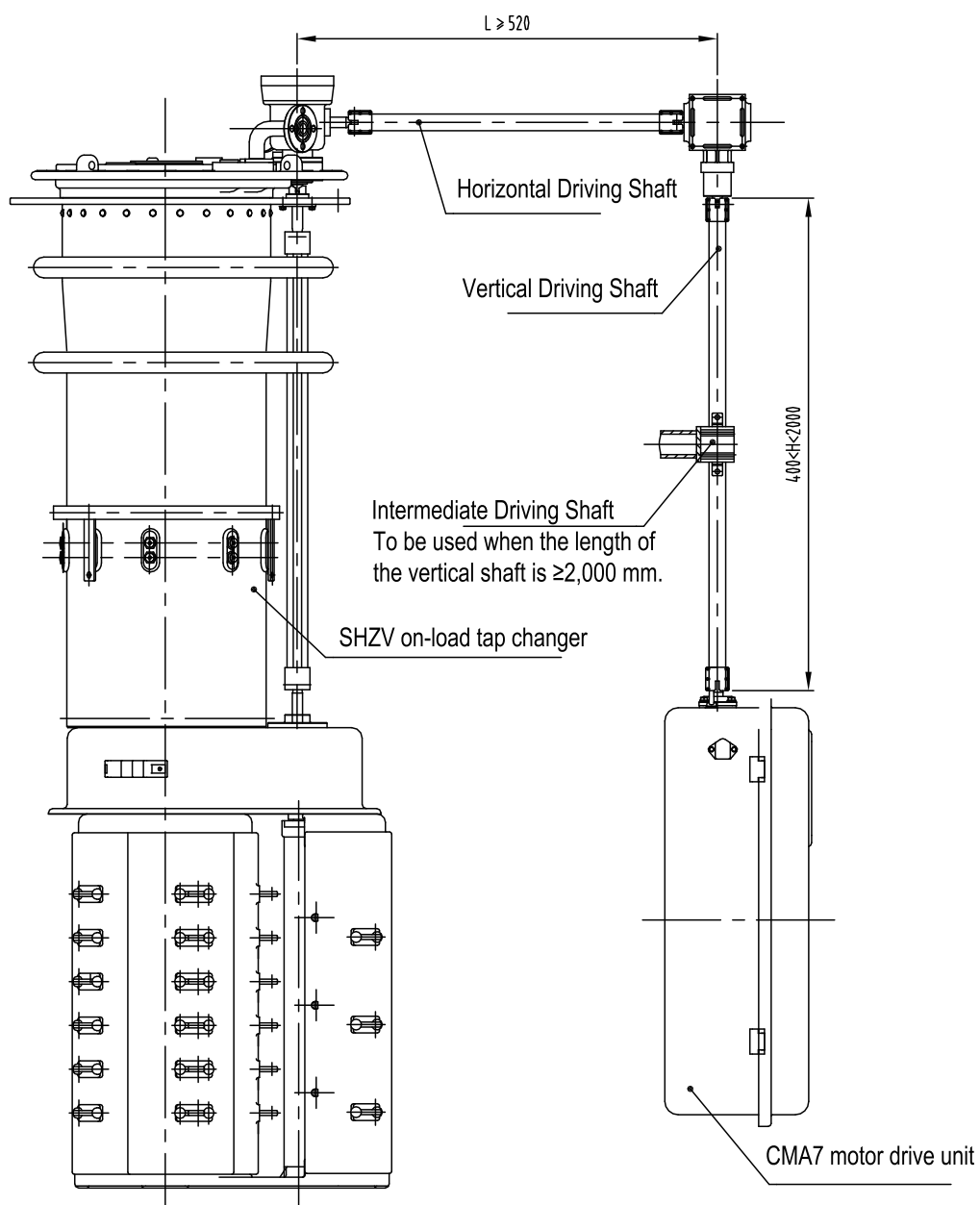


Shortened to Act as a Diverter Switch Handle for Tap Changer Operation
Extended to Act as Transformer Oil Release Valve Operating Rod

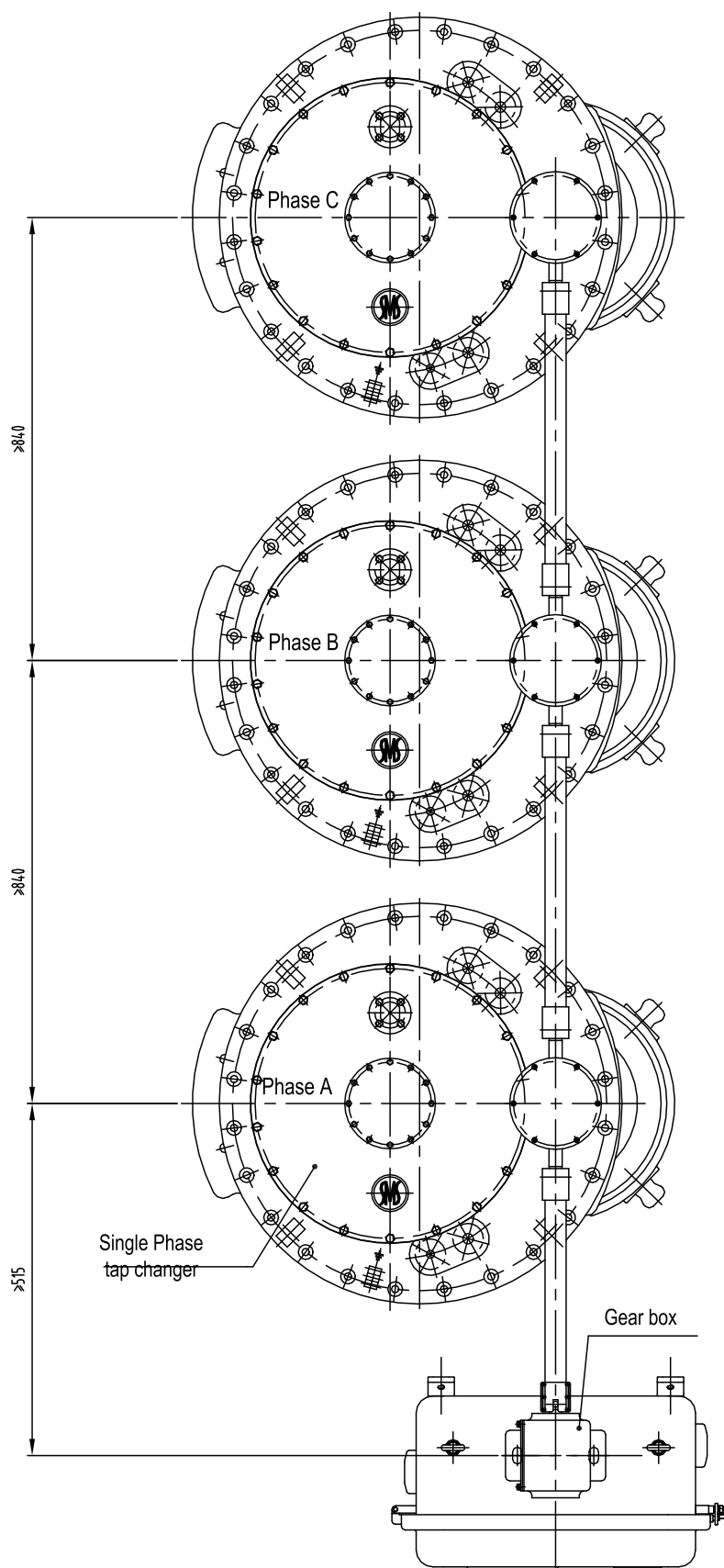
Appendix 44 Schematic drawing for connecting of SHZV OLTC and SHM



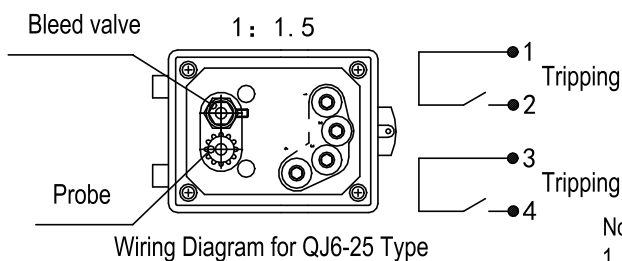
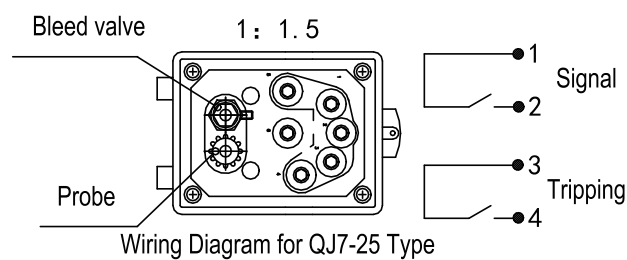
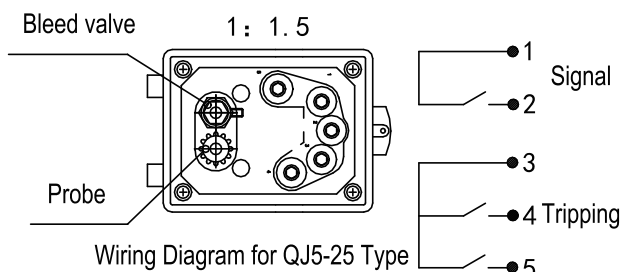
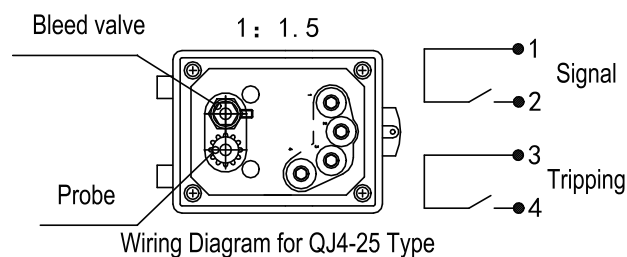
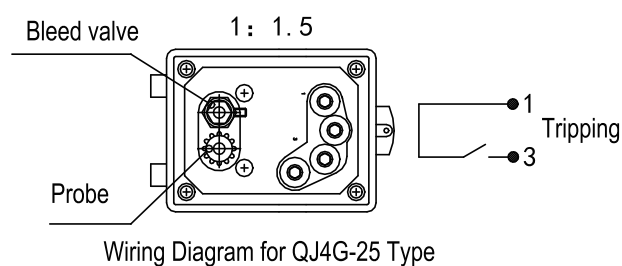
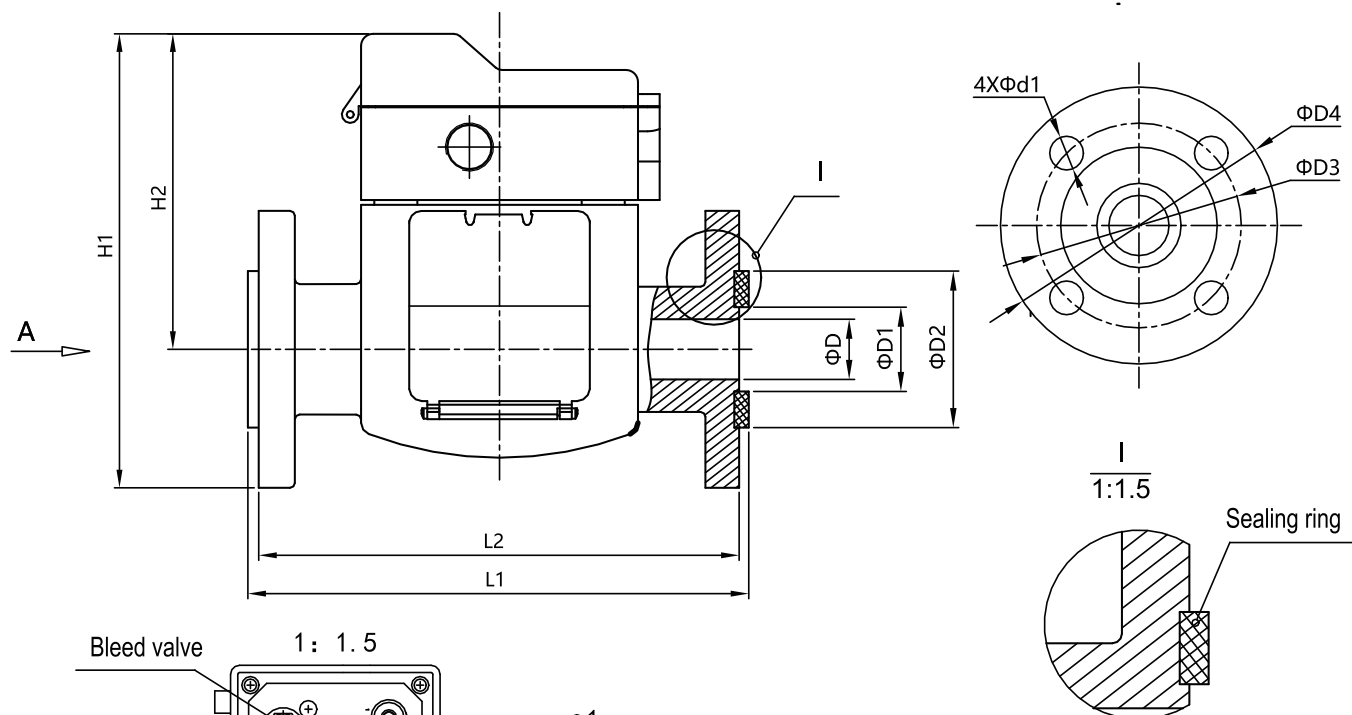
Appendix 45 Schematic drawing for connecting of SHZV OLTC and CMA7



Appendix 46 Schematic drawing for 3 units of single-phase SHZV connection arrangement



Appendix 47 Protective relay, overall dimension



Note:

1. Nylon Cable Fixing Head MA20-10 (M20x1.5 External Thread)
2. Probe is the Test Button

Model	D	D1	D2	D3	D4	d1	H1	H2	L1	L2	Remarks
QJ4-25	25	35	65	85	115	14	215	158	204	200	Single Signal, Single Jump Coil
QJ4G-25											Single Jump Coil
QJ5-25											Single Signal, Dual Jump Coil with Common Point
QJ6-25											Dual Independent Jump Coils
QJ7-25											Single Signal, Dual Independent Jump Coils



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